

The spherical magnetization law for the three-dimensional quantum Heisenberg ferromagnet

OpenAI

October 5, 2026

Abstract

We prove the spherical magnetization law for the three-dimensional nearest-neighbor isotropic quantum Heisenberg ferromagnet at every fixed positive quantum spin and every sufficiently low fixed positive temperature. As the even periodic cubes grow, the symmetric zero-field magnetization converges in moments to a uniform direction with a deterministic positive magnitude. This magnitude equals the right derivative at zero field of the infinite-volume pressure. The law is expressed through self-adjoint linear combinations of the spin components and requires no joint measurement of noncommuting observables.

Contents

1	Introduction	1
2	Exchange cycles, pins, and zero-free traces	4
3	Sparse pins on the torus	11
4	Spatial comparison at a vanishing field	20
5	Removing the field and comparing spatial scales	30
6	Equilibrium states and vanishing replica cross terms	34
7	Pressure tangents and the equilibrium magnetization radius	37
8	The total-spin limit	44

1 Introduction

An isotropic ferromagnet has no preferred direction in zero field. In a finite periodic volume, its Gibbs state is therefore rotation invariant even at temperatures where magnetized equilibrium states exist. The natural description of the ordered phase is a magnetization of fixed length and uniformly distributed direction. Rotation invariance alone does not establish this description: it allows a mixture of different lengths. The question addressed here is whether the symmetric finite-volume state selects a single length, and whether that length is the magnetization selected by an infinitesimal field.

1.1 Model and result

Fix $S \in \{\frac{1}{2}, 1, \frac{3}{2}, \dots\}$. Let S^1, S^2, S^3 be the irreducible spin- S matrices, normalized by

$$[S^a, S^b] = i \sum_c \epsilon_{abc} S^c, \quad \sum_{a=1}^3 (S^a)^2 = S(S+1)I.$$

For even $L \geq 4$, write $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$ and $V = L^3$. On $\bigotimes_{x \in \Lambda_L} \mathbb{C}^{2S+1}$, consider

$$H_{S,L} = - \sum_{\{x,y\} \text{ nearest neighbors}} \sum_{a=1}^3 S_x^a S_y^a, \quad M_L^a = \sum_{x \in \Lambda_L} S_x^a, \quad (1.1)$$

where each unordered bond is counted once. We use the Gibbs expectation

$$\langle A \rangle_{\beta,L} = \frac{\text{Tr}(A e^{-\beta H_{S,L}})}{\text{Tr} e^{-\beta H_{S,L}}}.$$

The pressure and its right derivative at zero field are

$$p_{S,\beta}(h) = \lim_{\substack{L \rightarrow \infty \\ L \text{ even}}} \frac{1}{\beta V} \log \text{Tr} e^{-\beta(H_{S,L} - h M_L^3)}, \quad (1.2)$$

$$m_{S,\beta} = p'_{S,\beta}(0+). \quad (1.3)$$

The pressure limit exists for this bounded finite-range interaction, and convexity gives the right derivative. The volume limit in (1.2) precedes removal of the field.

Theorem 1.1 (Spherical magnetization law). *For each fixed S there is a finite $\beta_0(S)$ such that, for every fixed finite $\beta \geq \beta_0(S)$, one has $m = m_{S,\beta} > 0$ and*

$$\lim_{\substack{L \rightarrow \infty \\ L \text{ even}}} \left\langle \exp \left(\frac{1}{V} \sum_{a=1}^3 t_a M_L^a \right) \right\rangle_{\beta,L} = \frac{1}{4\pi} \int_{\mathbb{S}^2} e^{m t \cdot n} d\sigma(n) = \frac{\sinh(m|t|)}{m|t|} \quad (t \in \mathbb{R}^3). \quad (1.4)$$

Here σ is surface-area measure on $\mathbb{S}^2$, and the quotient at $t = 0$ is interpreted as 1. In addition,

$$\lim_{\substack{L \rightarrow \infty \\ L \text{ even}}} \frac{1}{V^2} \sum_{x,y \in \Lambda_L} \langle S_x \cdot S_y \rangle_{\beta,L} = m^2. \quad (1.5)$$

The statement uses self-adjoint linear combinations of the spin components. It requires no joint finite-volume measurement of these noncommuting operators. Both the spin and the positive temperature remain fixed throughout the volume limit.

1.2 Context and antecedents

The dimensional distinction in this problem is fundamental. Mermin and Wagner [12] excluded spontaneous ferro- or antiferromagnetic order at positive temperature for finite-range isotropic Heisenberg models in dimensions one and two. In higher dimensions, infrared bounds proved continuous-symmetry breaking for classical models [7] and ordering for quantum antiferromagnets [6]. These quantum antiferromagnetic results do not supply the isotropic ferromagnetic ordering input used here. Another line of work establishes the low-temperature free-energy asymptotic predicted by spin-wave theory [5]. That asymptotic concerns a different limit from the field derivative or the zero-field magnetization distribution at a fixed positive temperature.

The spherical distribution itself has clear antecedents. Ueltschi’s description of three-dimensional loop models [23, Section 8.1] derives the transform $\sinh(hm)/(hm)$ for the spin- $\frac{1}{2}$ ferromagnet conditional on a conjectured Poisson–Dirichlet law for macroscopic loops. Björnberg, Fröhlich, and Ueltschi [2, Theorem 2.1] prove the spherical transform for the complete-graph quantum Heisenberg model at arbitrary fixed spin. Our setting has nearest-neighbor interactions on the three-dimensional lattice. We prove its magnetization law without assuming a distribution for macroscopic loop lengths.

The existence of spontaneous magnetization and the identification of its zero-field distribution are distinct questions. The companion manuscript [15] constructs magnetized equilibrium states for the isotropic nearest-neighbor quantum Heisenberg ferromagnet in dimensions at least three, for every fixed spin at sufficiently low positive temperatures. The companion [14] establishes the low-temperature asymptotic of the pressure-defined magnetization for nonnegative symmetric finite-range interactions whose support generates $\mathbb{Z}^3$. We use finite exchange and geometric estimates from these works. The present argument adds spatial control on periodic tori and a pressure argument excluding translation-ergodic equilibrium states with different magnetization magnitudes.

The exchange representation builds on permutation and random-walk representations of quantum spins [17, 4], Tóth’s random-stirring representation [21], and its higher-spin formulation through symmetric slots [13]. The finite pin estimates used below draw on the stability and negative dependence theory of Borcea, Brändén, and Liggett [3]. Our zero-free matrix products use the contraction principle associated with Asano’s and Ruelle’s developments of Lee–Yang theory [1, 18, 19]; the contraction needed here is proved explicitly. Mean entropy, equilibrium states, and pressure tangents are treated within the variational framework of Lanford and Robinson [10, 11]. We recall the required facts and prove the additional replica and harmonic-selection statements.

1.3 The two parts of the proof

Here is the obstruction that the proof must overcome. Let J_L denote the total-spin label in the rotation-invariant Gibbs state. Conditional on J_L , any component of M_L is uniform on $-J_L, -J_L + 1, \dots, J_L$. A fixed positive field and the definition of m already imply

$$\mathbb{P}(J_L/V > m + \epsilon) \longrightarrow 0 \quad (\epsilon > 0). \quad (1.6)$$

Thus it remains to prove

$$\liminf_{L \rightarrow \infty} \frac{\mathbb{E}[J_L(J_L + 1)]}{V^2} \geq m^2. \quad (1.7)$$

Together these bounds force $J_L/V \rightarrow m$ in probability and give (1.4). Section 8 supplies this last representation-theoretic argument in detail.

Two facts produce the lower bound. First, the squared magnetization averaged on large local boxes is controlled by the squared magnetization averaged on the whole torus, with an error vanishing as the local scale grows. This prevents local magnetization from canceling through spatial variation on larger scales. Second, every translation-ergodic equilibrium state has magnetization vector of length m . Here an equilibrium state maximizes mean entropy minus β times the physical energy density among translation-invariant states; an ergodic state is an extreme point of that invariant state space. Their variational and averaging properties are recalled in Section 6. The second fact identifies the local magnetization length to which the first fact applies.

Both facts use the same weak exponential estimate. To describe it, let $\mathcal{P}$ be a rectangular partition whose cell sides are comparable to a large scale $R \leq L$. For a deterministic site function

f with $|f_x| \leq 1$ and $\sum_{x \in C} f_x = 0$ in each cell, put $D_f = \sum_x f_x S_x^3$. We prove, for $b_R \asymp R^{-5/2}$ and a fixed $\kappa > 0$, that

$$\log \langle e^{\theta_R D_f} \rangle_{\beta, L} \leq C_\beta b_R V, \quad \theta_R = \kappa b_R \log R. \quad (1.8)$$

In particular the left side divided by $\theta_R V$ tends to zero uniformly in these tests. In nested rectangular partitions, apply this bound to signs of differences between a cell average and its parent average. A union bound and a telescoping identity then control all spatial scales. Applied to a product of two local zero-sum spin tests in different replicas, it forces the expectation of that product to vanish in every equilibrium state of the replicated system: finite-volume restrictions of such a state have relative entropy per site tending to zero with respect to the product Gibbs state.

Obtaining (1.8) is the spatial part of the proof. Sections 2–4 use exchange cycles and a field of strength b_R in a perpendicular direction to bound $R^{-5/2} V^{-1} \langle D_f^2 \rangle_{b_R}$ by a negative power of R . The field gives a killed one-particle kernel an invertible resolvent. A sparse-pin estimate controls the holes left after exposing unpinned cycles; a comparison on this random set then uses the zero sum of f in each cell. Section 5 turns the resulting variance bound into (1.8) through a zero-free complex strip.

The remaining argument identifies the magnitude. Negative squares of large-box magnetization averages would distinguish states of different lengths, but the corresponding perturbed Gibbs operators do not directly fit the exchange zero-freeness argument. Section 7 realizes these perturbations as first-order averages of random products of real matrices acting on the $2S$ spin- $\frac{1}{2}$ slots representing each site. Let $Z(z)$ be the normalized trace of such a product with complex field z . For every realization, $Z(z)$ is nonzero on $\operatorname{Re} z > 0$. Shared Gaussian variables couple its moments to replicated pressures; the zero-sum cross terms just described disappear from their pressure derivatives. Moment bounds then select harmonic limits of normalized $\log |Z(z)|$. Analyticity of their restrictions to the real axis excludes coexistence of different ergodic magnetization lengths. More concretely, an ergodic state of length a , oriented along the auxiliary field, contributes the affine function $xa - a^2$ to a limiting pressure tangent. Large positive x selects the branch $xm - m^2$; analytic continuation of that branch cannot dominate a competing branch of smaller length at every positive x . The spatial estimate, the replica argument, and the passage from random zero-free functions to harmonic pressure tangents are the main additional constructions of this paper.

1.4 Conventions

We suppress S in expectations and usually suppress β, L when no confusion is possible. Write X, Y, Z for the site spin components 1, 2, 3, in the standard representation where X, Z are real and Y is imaginary. Set $H_0 = -\beta H_{S, L}$. Constants c, C may depend on the fixed spin and on fixed geometric choices; C_β may also depend on the fixed inverse temperature. We first choose a sufficiently large finite β , and only then choose lower thresholds for L and spatial scales. All torus volume limits are through even L .

The finite exchange inputs are developed in Section 2, and the deterministic lattice estimates imported from the companions are stated with their hypotheses where they are used. Positivity of m follows directly from our torus sparse-pin estimate in Corollary 3.6; no sharp low-temperature asymptotic is needed. We increase $\beta_0(S)$ as needed below.

2 Exchange cycles, pins, and zero-free traces

The exchange representation gives two finite-dimensional tools for the proof. Pinning and exposing cycles turns a return probability into the first moment of a Markov hitting kernel. Stability of the

same exchange operators also gives zero-free traces after insertion of suitable single-slot matrices. We develop both tools here, keeping track of the conditioning in the first and the trace contraction in the second.

2.1 Symmetric slots and permutation layers

Put $\ell = 2S$ and let $\mathcal{I} = \Lambda_L \times \{1, \dots, \ell\}$ be the set of $N = \ell V$ slots. On $(\mathbb{C}^2)^{\otimes \mathcal{I}}$ write s_i^a for the Pauli matrices divided by two, and let

$$P_* = \prod_{x \in \Lambda_L} \frac{1}{\ell!} \sum_{\pi \in \mathfrak{S}_\ell} \mathcal{U}_{\pi, x}$$

be the projection onto the tensor product of the symmetric site subspaces. Here $\mathcal{U}_{\pi, x}$ permutes the slots at x . On this subspace $S_x^a = \sum_{i: x_i=x} s_i^a$ is the spin- S representation. In particular, if compression is needed for an individual slot operator, then

$$P_* s_i^a P_* = \ell^{-1} S_{x_i}^a P_*. \quad (2.1)$$

For each site bond use all ℓ^2 slot edges above it. On these edges put independent Poisson exchanges of rate $1/2$ during $[0, \beta)$, and at the seam from β to 0 insert an independent uniform slot permutation at every site. Let π be the resulting permutation of $\mathcal{I}$ and $\mathbb{P}_b$ its underlying picture law. If $\mathcal{E}$ is the set of slot edges, then

$$e^{H_0} P_* = e^{\beta |\mathcal{E}|/4} \mathbb{E}_b \mathcal{U}_\pi, \quad H_0 = -\beta H_{S,L}. \quad (2.2)$$

On the left H_0 is lifted to the slot space before compression. Indeed, $\mathbf{s}_i \cdot \mathbf{s}_j = \frac{1}{2} \mathcal{U}_{(ij)} - \frac{1}{4} I$, and the mean chronological Poisson product has generator $\frac{1}{2} \sum_{e \in \mathcal{E}} (\mathcal{U}_e - I)$. The all-slot sum above each bond commutes with P_* , while the seam average inserts P_* . This proves (2.2). This is Tóth's exchange representation [21], with the higher-spin construction of Nachtergaele [13, Section 2.2]; see also [14, Proposition 2.1].

If A is a polynomial in the physical site spins, let $\hat{A}$ be its slot lift obtained by replacing S_x^a by $\sum_{i: x_i=x} s_i^a$. The physical trace is then

$$\mathrm{Tr}_{\mathrm{phys}}(A e^{H_0}) = \mathrm{Tr}_{\mathrm{slot}}(\hat{A} e^{H_0} P_*). \quad (2.3)$$

Following lines through repetitions of one sampled picture gives its cycles. In this slot trace with diagonal spin insertions a cycle has a constant up or down color. Without insertions its contribution is two. We take slot-time points immediately after an instantaneous operation, including the seam. In finite volume there are almost surely finitely many marks, with distinct times avoiding any prescribed finite set of cuts.

We call a permutation law on a finite set *admissible* if it is a distributional limit of finite products of independent Bernoulli transpositions and deterministic permutations. Products here are ordered; no commutation is assumed. This class includes finite-time Poisson exchanges at arbitrary nonnegative rates, uniform permutations on specified subsets, and independent compositions of these operations. For the Poisson assertion, divide time into small intervals and apply each edge's Bernoulli indicator of at least one mark in a fixed order. The approximation agrees with the picture unless an interval contains at least two marks, whose probability tends to zero. For a uniform permutation, run positive-rate exchanges on every pair of the given subset and let time tend to infinity. This irreducible random walk on the finite permutation group converges to its uniform invariant law.

We use the following finite stability facts. A nonzero polynomial is *stable* if it does not vanish when all variables have positive imaginary parts. For every admissible law of a permutation π of a finite set I ,

$$\mathbb{E} \prod_{i \in I} (u_i + v_{\pi(i)}) \quad \text{is stable.} \quad (2.4)$$

Identification of variables preserves stability; differentiation, real specialization, and bounded-degree coefficientwise limits preserve it up to the zero polynomial. Finally, if a random subset $A \subseteq I$ has a stable generating polynomial $G_A(\mathbf{z}) = \mathbb{E} \prod_{j \in A} z_j$, then for every $J \subseteq I$,

$$\Pr(J \subseteq A) \leq \prod_{j \in J} \Pr(j \in A). \quad (2.5)$$

These are the partial-symmetrization and negative-dependence facts in [3, Theorems 4.20 and 4.9]; the exact finite formulations and their elementary proofs appear in [15, Lemmas 3.2–3.6]. In particular, the facts apply on a disjoint union of time layers, without any spatial assumptions.

2.2 Pins and exposure

The finite constructions in this subsection allow independent Poisson exchanges on an arbitrary finite slot graph, together with finitely many prescribed deterministic permutations, Bernoulli transpositions, and uniform permutations on prescribed subsets at fixed times. All random operations are independent under the base law. Their order at coincident prescribed times is fixed, and a slot-time point at such a time lies after the entire ordered bundle of operations. Thus extra stationary lines and extra instantaneous exchanges are included.

A *pin* is a specified slot-time point required to have up color. For a finite deterministic pin set E , let n_E be the number of cycles avoiding E , let $\mathcal{B}_E$ be their union, and define

$$\mathcal{Z}_E = \mathbb{E}_b 2^{n_E}, \quad \mathbb{P}_E(d\omega) = \mathcal{Z}_E^{-1} 2^{n_E(\omega)} \mathbb{P}_b(d\omega). \quad (2.6)$$

Cycles meeting pins are up; the others are colored independently and fairly. For a nonempty single-time set T disjoint from E , put $P = E \cup T$ and let σ take each pin in P to the next pin on its directed cycle, at strictly positive elapsed time. Set

$$r(E, T) = \frac{1}{|T|} \sum_{i \in T} \mathbb{P}_P\{\sigma(i) \in T\}. \quad (2.7)$$

The same sampled picture is repeated in defining σ .

To describe exposure, label lines at time zero and reveal all labels on cycles missing P , their trajectories through a period, and all their endpoint images. The trajectories record their positions before and after each elementary operation, including intermediate stages of an ordered bundle at one time. Denote these data by $\mathbf{a}$. The slots not occupied by exposed labels are called *holes*. For fixed exposure, define a fresh law $\mathbb{Q}_{\mathbf{a}}$ on the holes as follows. Exchanges whose two endpoints are holes have their original independent Poisson clocks. At a transposition exchanging an exposed label with a hole, that hole moves into the vacated slot. These forced transports are deterministic bijections between successive hole layers. A partly revealed uniform permutation is completed by a uniform bijection between the remaining source and target slots; a prescribed deterministic permutation restricts to a deterministic bijection between those hole layers. An elementary Bernoulli transposition touching an exposed label has its outcome revealed; one touching only holes retains its independent Bernoulli law. Let G_P denote the event that every cycle of this fresh picture meets P .

Lemma 2.1 (Finite pin inputs). *For the finite picture above the following assertions hold.*

(i) *For every finite E and nonempty single-time T disjoint from E ,*

$$\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq (2r(E, T))^{|T|}. \quad (2.8)$$

(ii) *Almost surely under $\mathbb{P}_P$, $g_P(\mathbf{a}) := \mathbb{Q}_{\mathbf{a}}(G_P) > 0$, and the conditional law of the remaining picture is $\mathbb{Q}_{\mathbf{a}}(\cdot \mid G_P)$.*

(iii) *Let $\omega_{\mathbf{m}}$ be the Poisson mark counting measure and let $\lambda_{\mathbf{b}}$ be edge rate times Lebesgue time measure. For every integer $j \geq 1$ and nonnegative measurable function F of j mark locations,*

$$\mathbb{E}_E \sum_{(u_1, \dots, u_j) \in \omega_{\mathbf{m}, \neq}^j} F(u_1, \dots, u_j) \leq 2^j \int F d\lambda_{\mathbf{b}}^{\otimes j}. \quad (2.9)$$

The sum is over ordered distinct marks, and may be restricted to any prescribed time order. Independent additional fresh Poisson clocks may be included with their own, undoubled, factorial intensity.

Proof. The statements are the finite-pin forms of [14, Proposition 2.4, Proposition 2.5, and Lemma 2.3]. We give the arguments to specify the scope of the extra operations and the measure used after exposure.

Cut at all pin times and at zero, and let $\mathcal{V}$ be the disjoint union of the slot layers at these cuts. The maps between successive cuts are independent admissible bijections. Their symbols in disjoint input and output variables multiply to give a stable polynomial $\mathbb{E}_{\mathbf{b}} \prod_{i \in \mathcal{V}} (x_i + y_{\pi(i)})$. Delete a nonpin q by setting $x_q = y_q = t$ and extracting the coefficient of t . If its predecessor and successor are i, j , then $(x_i + t)(t + y_j)$ is replaced by $x_i + y_j$. A fixed point instead contributes $2t$, hence a factor two. Successive deletion therefore gives the nonzero stable polynomial

$$\mathbb{E}_{\mathbf{b}} \left[2^{n_P} \prod_{i \in P} (x_i + y_{\sigma(i)}) \right].$$

Differentiate in the input variables indexed by $P \setminus T$, set all input variables to zero, and divide by $\mathcal{Z}_P$. The result is $\mathbb{E}_P \prod_{i \in T} y_{\sigma(i)}$, the stable generating polynomial of the subset $\sigma(T)$. By (2.5) and the arithmetic-geometric mean inequality,

$$\mathbb{P}_P\{\sigma(T) = T\} \leq \prod_{j \in T} \mathbb{P}_P\{j \in \sigma(T)\} \leq r(E, T)^{|T|}.$$

Here $\sum_{j \in T} \mathbb{P}_P\{j \in \sigma(T)\} = |T|r(E, T)$ by bijectivity. The event $\sigma(T) = T$ is precisely $T \subseteq \mathcal{B}_E$: iterating the next-pin permutation from T then meets no pin in E . Finally, $2^{n_P} \leq 2^{n_E} \leq 2^{|T|} 2^{n_P}$ and $\mathcal{Z}_E \geq \mathcal{Z}_P$, proving (i).

For (ii), first fix a proposed set A of initial exposed labels. Use two independent Poisson clock families on every edge, each with the original rate: accept the first family's clock when an endpoint contains an A -label and the second when neither does. This realizes the base process. Given the past, the superposed clock has twice the total original rate, each edge is accepted with its original relative rate, and the rejection probability is $1/2$. The A -label paths use only the first family. Given these paths, the second family restricted to the hole intervals remains fresh. Revealing the A -label images at a uniform permutation leaves a uniform residual bijection. At a Bernoulli

transposition its outcome is revealed exactly when an exposed label is involved; other outcomes remain independent. This constructs $\mathbb{Q}_{\mathbf{a}}$.

More explicitly, let μ_A be the base law of the paths and endpoint images of the proposed labels. Let F_A mean that those paths avoid P and close on A , and let c_A count their cycles on this event. For every bounded measurable function Φ of the exposure and remaining picture,

$$\mathbb{E}_P \Phi = \mathcal{Z}_P^{-1} \sum_A \int_{F_A} 2^{c_A(\mathbf{a})} \mathbb{E}_{\mathbb{Q}_{\mathbf{a}}} [\mathbf{1}_{G_P} \Phi] \mu_A(d\mathbf{a}). \quad (2.10)$$

Each full picture occurs in exactly one branch, namely its set of labels on cycles missing P . The exposure marginal on that branch is therefore $\mathcal{Z}_P^{-1} 2^{c_A} g_P \mu_A$ restricted to F_A . It assigns zero mass to $g_P = 0$, and division by g_P proves (ii). In particular the exposure marginal includes the all-hit probability g_P ; it is not the unweighted law of the proposed histories.

For (iii), inserting one transposition joins two cycles or splits one. It increases the number of cycles missing the fixed pins by at most one, and so multiplies the weight by at most two, regardless of the other instantaneous operations. The Poisson insertion formula for j ordered distinct marks, followed by division by $\mathcal{Z}_E$, gives (2.9). Independent fresh clocks have their ordinary factorial measure, proving the last assertion. $\square$

2.3 The conditional next-pin first moment

The conditioning in Lemma 2.1(ii) couples entire cycles. The next result explains exactly why its next-pin first moment can nevertheless be computed by a walk with fresh transitions. Matrices in this subsection use the column-forward convention: $\Pi e_i = e_{\pi(i)}$.

Lemma 2.2 (Conditional next-pin kernel). *Let Π have an admissible permutation law on a finite set $\mathcal{V} = I \sqcup D$, and put $K = \mathbb{E}\Pi$. Let G_I be the event that every permutation cycle meets I . On G_I , let Σ be the permutation matrix on I taking each pin to its next positive visit to I . Then*

$$\Pr(G_I) = \det(I_D - K_{DD}). \quad (2.11)$$

For every complex matrix W on I , there is the polynomial identity

$$\mathbb{E}[\mathbf{1}_{G_I} \det(I_I - W\Sigma)] = \det \begin{pmatrix} I_I - WK_{II} & -WK_{ID} \\ -K_{DI} & I_D - K_{DD} \end{pmatrix}. \quad (2.12)$$

If $\Pr(G_I) > 0$ and $I \neq \emptyset$, then K_{DD} is transient and

$$\mathbb{E}[\Sigma \mid G_I] = K_{II} + K_{ID}(I_D - K_{DD})^{-1}K_{DI}. \quad (2.13)$$

This is the first positive hitting kernel of I for the Markov chain with independent transitions K . Empty determinants are one and empty correction terms are zero. In particular (2.12), and its linear coefficients in W , remain valid when the all-hit probability is zero.

Proof. We reproduce the finite argument in [14, Lemma 2.6 and Proposition 2.7], equivalently [15, Lemma 4.2 and Proposition 4.3]. Write $\bigwedge^k M$ for the action of M on alternating k -tensors; its entries are the k -rowed minors of M . For $0 \leq k \leq |\mathcal{V}|$, exterior powers satisfy

$$\mathbb{E} \bigwedge^k \Pi = \bigwedge^k K. \quad (2.14)$$

Indeed, for a transposition T_{ij} , $T_{ij} - I = -(e_i - e_j)(e_i - e_j)^*$ has rank one. Every minor of $I + u(T_{ij} - I)$ is affine in u , so its exterior power is the mean exterior power of a Bernoulli transposition of probability u . Independence and multiplicativity of exterior powers extend this identity to products; continuity of minors extends it to their limits.

For any fixed matrix B , the principal-minor expansion now gives

$$\mathbb{E} \det(I - B\Pi) = \sum_k (-1)^k \text{Tr} \left[\left(\bigwedge^k B \right) \mathbb{E} \bigwedge^k \Pi \right] = \det(I - BK). \quad (2.15)$$

Take $B = \text{diag}(W, I_D)$. If a permutation cycle is contained in D , its indicator is a fixed vector of $B\Pi$, making the determinant zero. Otherwise Π_{DD} is nilpotent, and elimination of this block follows the successive nonpins on each cycle. It yields $\det(I - B\Pi) = \det(I_I - W\Sigma)$. This proves (2.12); setting $W = 0$ proves (2.11) without any invertibility assumption.

When the determinant is positive, take its Schur complement in (2.12). Replace W by tW and compare linear coefficients, using $\det(I - tWA) = 1 - t \text{Tr}(WA) + O(t^2)$. Since W is arbitrary, the resulting trace identities recover every entry in (2.13). The nonnegative matrix K_{DD} is substochastic. Its spectral radius is at most one; if it were one, the Perron eigenvalue would make $I_D - K_{DD}$ singular. Consequently

$$(I_D - K_{DD})^{-1} = \sum_{n \geq 0} K_{DD}^n.$$

The term K_{II} is a direct hit, and $K_{ID}K_{DD}^nK_{DI}$ is a hit after $n + 1$ intervening states in D . This proves the Markov interpretation. $\square$

Apply the lemma at a fixed exposure by cutting at all pin times and at the finitely many prescribed changes in hole sets. Identify the hole layers, which have equal cardinality. The unconditioned interval maps are independent admissible bijections; their direct sum followed by the cyclic shift of the layers is an admissible permutation of the layered set. Its mean matrix gives the hole walk: jumps at the original edge rates ($1/2$ for physical edges), prescribed deterministic restrictions and forced transports, fresh residual uniform choices at each prescribed uniform operation (including the seam), and fresh hole-only Bernoulli outcomes. Repeat this environment periodically, making independent transition choices on every passage, including later passages through the same physical time interval. If τ_T^+ is the first positive visit to T and τ_E the first visit to E , then

$$|T|r(E, T) = \mathbb{E}_P \sum_{i \in T} \sum_{\mathbf{a}, i}^{\text{walk}} \Pr(\tau_T^+ < \tau_E). \quad (2.16)$$

This follows by Lemma 2.2 under $\mathbb{Q}_{\mathbf{a}}(\cdot \mid G_P)$ and then Lemma 2.1(ii). The identity concerns a conditional first moment. The original repeated permutation paths retain their cycle dependence; it is the determinant identity that replaces their next-pin mean by fresh Markov transitions.

2.4 Disk contractions and trace zero-freeness

We next use (2.4) for complex matrix insertions. In any common qubit basis the polynomial of a permutation layer, evaluated between input kets $(1, x_i)^T$ and algebraic output covectors $(1, y_i)$, without complex conjugation, is

$$\mathbb{E} \prod_i (1 + x_i y_{\pi(i)}). \quad (2.17)$$

It has no zeros when all $|x_i|, |y_i| < 1$. To see this, substitute

$$u_i = i \frac{1 + x_i}{1 - x_i}, \quad v_j = i \frac{1 - y_j}{1 + y_j}$$

in (2.4). Both families are in the upper half-plane, and $u_i + v_{\pi(i)} = 2i(1 + x_i y_{\pi(i)}) / [(1 - x_i)(1 + y_{\pi(i)})]$; the product of denominators is independent of π .

Lemma 2.3 (Zero-free layered traces). *On finitely many qubit slots, take a finite cyclic product of permutation layers and tensor products of single-slot matrices, with at least one permutation layer. Each permutation layer is a nonzero scalar times the mean permutation operator of an admissible law. Between every two successive permutation layers, suppose that the single-slot matrix on every slot induces a projective map*

$$(1, x)^T \mapsto a(x)(1, \phi(x))^T$$

with $a(x) \neq 0$ and $|\phi(x)| < 1$ for all $|x| \leq 1$. Then the trace of the cyclic product is nonzero.

Proof. Ignore the nonzero scalar factors. Group each single-slot layer with the following permutation layer. By compactness, its slot maps send $|x| < 1 + \varepsilon$ into $|\phi(x)| < 1$, with nonvanishing $a(x)$, for some common $\varepsilon > 0$. Thus its multiaffine matrix polynomial is nonzero for $|x_i| < 1 + \varepsilon$ and $|y_i| < 1$, by (2.17). Multiply the layer polynomials in disjoint variables.

The trace is obtained by contracting each output variable with the input variable on the matching slot of the next layer. Such a contraction acts on a bivariate section by

$$A + Bx + Cy + Dxy \mapsto A + D. \quad (2.18)$$

To verify preservation of nonvanishing, fix all other variables in their prescribed disks. Choose $r_x < 1 + \varepsilon$ and $r_y < 1$ with $r_x r_y > 1$, and set $x = r_x z$, $y = r_y z$. The resulting polynomial has no zero in $|z| < 1$ and has nonzero constant term A . If its quadratic coefficient is nonzero, the product of its two roots has modulus at least one, giving $|D| r_x r_y \leq |A|$. If it has degree at most one, $D = 0$. In either case $|D| < |A|$ whenever $D \neq 0$, so $A + D \neq 0$. After each contraction the polynomial stays multiaffine and nonzero on the product of the remaining disks. Contracting all pairs proves the claim. This is a disk form of the contraction method of Asano and Ruelle [1, 18], applied here to the complete cyclic trace. $\square$

Choose now the s^2 eigenbasis with vectors $(1, \pm i)^T / \sqrt{2}$ in the s^3 coordinates. The following three slot maps will cover both the field comparison and the later pressure argument.

Lemma 2.4 (Single-slot maps). *In this basis, matrices in $SL_2(\mathbb{R})$, real in the s^3 coordinates, act by automorphisms of the unit disk. If $\operatorname{Re} z > 0$, the matrix e^{zs^2} sends the closed disk strictly into the open disk. There are absolute constants $b_*, a_0 > 0$ such that, whenever $0 < b \leq b_*$, $f \in [-1, 1]$, and $|\operatorname{Im} z| < a_0$, the matrix*

$$e^{bs^2/2} e^{bzf s^3} e^{bs^2/2} \quad (2.19)$$

also sends the closed disk strictly into the open disk, with its projective coordinate defined throughout the closed disk.

Proof. The projective coordinate in the s^3 basis is $\zeta = i(1 - x)/(1 + x)$. It identifies the unit disk with the upper half-plane. A real determinant-one matrix acts there by an upper-half-plane automorphism, proving the first assertion. In the s^2 basis e^{zs^2} acts by $x \mapsto e^{-z}x$, proving the second.

For the last assertion, the rightmost factor first gives $|x| \leq r = e^{-b/2}$. Its upper-half-plane image satisfies

$$\sin(\arg \zeta) = \frac{\operatorname{Im} \zeta}{|\zeta|} = \frac{1 - |x|^2}{|1 - x||1 + x|} \geq \frac{1 - r^2}{1 + r^2} = \tanh(b/2).$$

For $0 < b \leq b_*$, this is at least cb with an absolute $c > 0$. Thus $\arg \zeta$ stays at angular distance at least cb from both 0 and π . The middle factor sends ζ to $e^{-bz} \zeta$, changing its argument by at most $b|\operatorname{Im} z|$. Choose $a_0 < c$; the image is still in the upper half-plane. The final factor shrinks the resulting disk coordinate by $e^{-b/2}$. None of these maps has a pole on the compact set being mapped, proving the stated strictness and definedness. $\square$

The next section uses a small positive field to supply pins. The finite identities proved here reduce uniform bounds on their avoidance to estimates for the corresponding hole walk.

3 Sparse pins on the torus

We now show that a sparse collection of prescribed slot points is unlikely to lie on cycles avoiding the field pins. This will supply the geometric control of available slots used in Section 4. The induction follows the pinned-boundary argument of [14, Section 4], itself developed from [15, Sections 5 and 7]. On the torus there is no pinned boundary: a small uniform field supplies the killing, and cutting off flows to infinity leaves an error that this killing must absorb.

3.1 Soft pins and the ghost representation

Rotate coordinates so that the field selects the up color. If $q \in (0, 1)$, $w = 1 - q = e^{-b}$, its single-slot factor, after removing a scalar, is

$$\operatorname{diag}(1, w) = wI + qP_{\text{up}}. \quad (3.1)$$

Thus we may independently select a pin on each real slot at the cut with probability q , and then weight the exchange picture by 2^c , where c counts the cycles missing all pins. Pinned cycles have the up color; the other cycles receive independent fair colors. Expanding (3.1) proves that the resulting spin law is the Gibbs law with exponent $H_0 + b \sum_x Y_x$, after rotating the up axis back to Y .

For the induction we need a fixed pin set. Add one stationary *ghost slot* for each real slot, pin each ghost at time $\beta/2$, and, just after the seam at time zero, independently transpose each real slot with its ghost with probability q . The time-zero layer is taken after these switches. A selected switch inserts the corresponding ghost line into the real cycle; that cycle then contains a ghost pin. An unselected ghost remains a separate pinned cycle and contributes weight one. Consequently every switch configuration has exactly the weight of the corresponding soft-pin configuration. This remains true with any finite set of deterministic pins on real slot points. Figure 1 illustrates the two cases.

Let E denote all the ghost pins together with any finite set of deterministic real-slot pins. We write $\mathbb{P}_E$ for the normalized picture law with density proportional to 2^{c_E} , and $\mathcal{B}_E$ for the slot-time points on cycles missing E . These cycles contain only real slots. At coincident prescribed operations, a slot point means the point just after the entire ordered bundle, as in Section 2. Distances between sites are wrapped supremum distances, denoted d_L ; balls are denoted by $B_h(z)$.

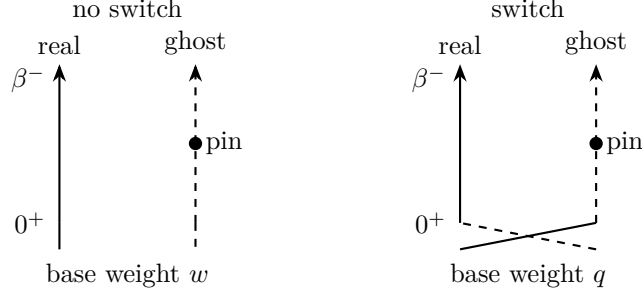


Figure 1: One real slot and its stationary ghost, shown on a cut-open time circle. An absent switch leaves the ghost as a separate pinned cycle. A present switch inserts its pinned line into the real cycle. In the unconditioned fresh real-hole traversal killed on entry into a ghost, passage through this layer survives with probability $w = 1 - q$. The rest of the real exchange picture is suppressed.

Fix $\alpha = 1/100$ and a sufficiently large constant $C_s \geq 6$. A finite collection of real slot points is *sparse* if its site coordinates, counted with multiplicity, satisfy

$$\#\{i : x_i \in B_h(z)\} \leq C_s(1 + h)^\alpha \quad (z \in \Lambda_L, h \geq 0). \quad (3.2)$$

The same definition applies to lists of centers. In particular, different slots at the same site count separately. The freedom to choose C_s large will permit the finite collections of nearby test points used in Section 4.

Proposition 3.1 (Sparse-pin bound). *Fix S and C_s . There is $\beta_0 < \infty$ such that, for every fixed $\beta \geq \beta_0$, there is $L_*(\beta) < \infty$ with the following property. For every even $L \geq L_*(\beta)$, every q satisfying $L^{-5/2} \leq q < 1$, every pin set E as above, and every sparse set T of real slot points at one deterministic time,*

$$\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq p^{|T|}, \quad p = \beta^{-9/10}. \quad (3.3)$$

The constants and thresholds are independent of the number, locations, and times of the extra pins.

The proof is a finite downward induction on deterministic pins. Its larger-pin hypothesis gives regular hole graphs; diffusion and a flow-energy estimate then make next-pin returns rare enough to remove the test pins by Lemma 2.1.

If T is empty the assertion is immediate; if it meets E , the event is impossible. We henceforth consider nonempty $T \cap E = \emptyset$, put $P = E \cup T$, and expose the cycles missing P under $\mathbb{P}_P$. All other slot points are called *holes*. Lemmas 2.1 and 2.2 apply to this enlarged picture. Here are the details concerning ghosts.

For any proposed set of exposed time-zero real labels, reveal its trajectories, its seam images, and the switch outcomes encountered by those trajectories. Closure into cycles avoiding P requires every such switch to be absent. Separate Poisson clocks according to whether they touch the currently exposed labels, as in the exposure disintegration of Lemma 2.1. Conditional on the revealed histories, the unused clocks are fresh on pairs of holes; the remaining seam images are uniform completions within each site; and switches on the other real-ghost pairs are still independent Bernoulli switches. When an exposed label jumps through a hole, that hole is transported in the opposite direction. A mark between two exposed labels does not move a hole. Requiring these to be all the missing cycles conditions the fresh picture on every remaining cycle meeting P . The exposed cycles contribute only their fixed factor 2^{c_P} ; therefore this description gives

the conditional law, including its all-hit conditioning. That conditioning has positive probability at almost every exposure.

Every ghost is a hole and is pinned. Missing real cycles never use a switch, so the number of real holes is constant in time. The seam does not change their number at any site. In particular the factorial-mark estimate in Lemma 2.1 remains valid: inserting a real-edge transposition changes the number of missing cycles by at most one, even in the presence of switches.

By Lemma 2.2, conditional next-pin probabilities are first-hit probabilities for the unconditioned fresh walk, with fresh transitions on each repeated traversal. Delete the other real-pin stops and kill that walk on entering a ghost. This only increases the probability that its next pin belongs to T : after entering a ghost, the walk encounters its pin before its next opportunity to return to a real slot. The one-turn real-hole kernel at the time of T is

$$U = wK, \quad U_+ = U, \quad U_- = U^*, \quad Q = \frac{1}{2}(U + U^*), \quad E_y = \langle y, (I - Q)y \rangle \geq q\|y\|_2^2. \quad (3.4)$$

Here and throughout this section matrices are column-forward. The matrix K is doubly stochastic: its factors are symmetric hole-walk heat kernels, forced transport bijections, and uniform remaining seam completions. Exactly one switch layer is crossed per turn, and its real-to-real mean is wI , which proves $U = wK$. All factors, their adjoints, and spatially killed versions contract counting 1-, 2-, and supremum norms. For example, a nonnegative matrix whose row and column sums are at most one satisfies $\sum_i |(Av)_i|^2 \leq \sum_{i,j} A_{ij}|v_j|^2 \leq \|v\|_2^2$. This also proves the last inequality in (3.4).

3.2 The deterministic geometric input

We state precisely the infinite-lattice results that will be transferred to the torus. An available-slot graph on $\mathbb{Z}^3$ has between zero and ℓ slots at each site, all ℓ slots outside a finite set, and all pairs of available slots joined across every nearest-neighbor site bond. A site with no available slot is *blocked*. Join blocked sites at supremum distance at most eight and write $\mathcal{C}(w)$ for the component of a blocked site w . Fix $\nu = 1/300$. The constants M_* and A_0 below depend only on this lattice geometry and ℓ .

A graph is *regular at z with radius $A \geq A_0$* if

- (i) every ball $B_{M_*}(v)$, with $v \in B_{11A}(z)$, contains fewer than six blocked sites;
- (ii) each blocked $w \notin B_{10A}(z)$ satisfies $\max_{v \in \mathcal{C}(w)} |v - w|_\infty \leq (1 + |w - z|_\infty)^\nu$.

The anchor itself need not have an available slot. One may take $M_* \geq \max\{48, \rho(3, 44) + 2\}$, where the local-connectivity radius $\rho(3, 44)$ has the following property: two unblocked sites at distance at most 44 can be joined inside $B_{\rho(3, 44)}$ about the first site if that ball contains fewer than six blocked sites. The path has length at most $(2\rho(3, 44) + 1)^3$. This is [15, Lemma 5.5 and Equation (5.13)]; regularity is [14, Definition 3.2], equivalently [15, Definition 5.3]. Take $A_0 \geq M_* + 47$ large enough for the following two conclusions.

Lemma 3.2 (Lattice Nash inequality and unit flows). *In an available-slot graph regular at z with radius $A \geq A_0$, put*

$$\mathcal{D}(v) = \frac{1}{2} \sum_{\{i,j\} \text{ available edge}} (v_i - v_j)^2.$$

Every nonnegative v supported over $B_A(z)$ satisfies

$$\|v\|_2^{10/3} \leq C\mathcal{D}(v)\|v\|_1^{4/3}. \quad (3.5)$$

For every available slot a at $y \in B_A(z)$, there exists an oriented edge flow F_a , with divergence defined as outgoing minus incoming flow, such that

$$\operatorname{div} F_a = \mathbf{1}_{\{a\}}, \quad |F_a(e)| \leq C(1 + \operatorname{dist}(e, y))^{-k}, \quad k = 2 - 3\nu = 199/100. \quad (3.6)$$

Here $\operatorname{dist}(e, y)$ is the minimum supremum distance to the two endpoint sites. The constants are uniform over the graph, A , z , and the source. Sites with any number of available slots between one and ℓ are permitted.

These are [14, Proposition 3.3 and Lemma 3.4]; the underlying geometric constructions are [15, Propositions 5.6 and 6.2]. Their hypotheses are deterministic and impose no relation between A and a time scale. Only these stated conclusions of the infinite-lattice results will be used here.

At a given exposure and physical time, make such a graph for each torus anchor z : lift the real holes in $B_{\lfloor L/4 \rfloor}(z)$ to $\mathbb{Z}^3$, and declare every slot outside this box available. We say the slice is regular at z when this graph is regular. We shall use

$$A = \lceil C_{\text{loc}} \beta \rceil, \quad 11A + M_* < \lfloor L/4 \rfloor, \quad (3.7)$$

where C_{loc} is fixed sufficiently large below. Thus the Nash inequality for functions supported in $B_A(z)$ uses exactly the torus edges, with zero values beyond the support.

3.3 From grid pins to regular slices

Choose a finite circular grid $\mathcal{G}$ containing zero, with successive gaps at most p^{12} . For a set P of ghost pins and real pins on this grid, let $\mathbf{l}(P)$ assert (3.3) under $\mathbb{P}_P$ for every sparse test at one grid time. In this subsection we assume $\mathbf{l}(P)$. This is the hypothesis for a larger pin set in the eventual downward induction; no estimate for a smaller pin set is assumed.

At time t , write $\mathcal{O}_t$ for the sites without a real hole. At a grid time, simultaneous blocking of prescribed sparse sites costs at most p^n : choose one fixed slot at each site and apply $\mathbf{l}(P)$. If any chosen slot is pinned, the event is impossible. The same conclusion can therefore be used without checking disjointness separately.

Lemma 3.3 (Blocking and component tails). *Under $\mathbf{l}(P)$, at any deterministic time t ,*

$$\mathbb{P}_P(A' \subseteq \mathcal{O}_t) \leq Cp^6 \quad \text{for six prescribed distinct sites,} \quad (3.8)$$

$$\mathbb{P}_P(A' \subseteq \mathcal{O}_t) \leq (C\sqrt{p})^{|A'|} \quad \text{for a nonempty sparse set of sites.} \quad (3.9)$$

There are $c, C, \lambda > 0$, uniform for sufficiently large β , such that the blocked component formed with steps of supremum length at most eight satisfies

$$\mathbb{P}_P\{z \in \mathcal{O}_t, \max_{v \in \mathcal{C}_t(z)} d_L(z, v) \geq h\} \leq Ce^{-ch^\lambda} \quad (h \geq 1). \quad (3.10)$$

All constants are independent of the grid and its size.

Proof. Let s be the grid point preceding t , with $s = t$ if $t \in \mathcal{G}$. A site blocked at t either was blocked at s , or has an incident real-edge mark in $(s, t]$. The seam preserves hole counts at each site; the switch layer also preserves the real-hole set of an exposure, as observed above. The union of the stars of six sites has bounded total intensity. Factorial domination in Lemma 2.1 bounds the probability of a mark there by Cp^{12} , giving $p^6 + Cp^{12} \leq Cp^6$.

For n sparse sites, either at least $\lceil n/2 \rceil$ were blocked at s , costing at most $2^n p^{\lceil n/2 \rceil}$, or at least that many have incident marks. A mark touches at most two of the selected sites, so the second alternative requires $j = \lceil n/4 \rceil$ distinct marks. The union of the stars has intensity at most Cn ; hence its probability is at most

$$\frac{(Cnp^{12})^j}{j!} \leq (C'p^{12})^j,$$

using $j! \geq (j/e)^j$ and $n/j \leq 4$. Enlarging the constant gives (3.9), without an independence assumption between the two alternatives.

For the component tail, fix an integer $a > 100$ so large that $a^\alpha > 4$, and set $h_j = 16a^j$. A blocked path from z to wrapped distance h_j yields two smaller crossings: the first starts at z , and the second at the first path site at distance at least $h_j/2$ from z . Truncate each at its first reach of distance h_{j-1} from its own start. The starts are between $h_j/2$ and $h_j/2 + 8$ apart, and each truncated crossing stays within radius $h_{j-1} + 8$. Thus their mutual distance is at least $h_j/2 - 2(h_{j-1} + 8) \geq h_j/4$. The remaining portion of the original crossing has sufficient length for the second truncation, since its endpoint is at least $h_j/2 - 8$ from the second start. These statements use only the triangle inequality for wrapped distance.

Recursion gives an ordered binary witness with 2^j distinct blocked leaves. At every split of scale h_k the two descendant leaf sets are at distance at least $h_k/4$. Therefore a ball of radius less than $h_k/8$ meets at most one branch at each split of scale h_k or larger, and contains at most 2^{k-1} leaves. For $h_{k-1}/8 \leq h < h_k/8$, this is bounded by $C_s(1+h)^\alpha$, because $a^\alpha > 4$. Below $h_0/8$ there is at most one leaf, and at or above $h_j/8$ the same comparison allows all 2^j leaves. Hence every such witness is sparse, at every center and every radius.

At a split of scale h_k , the second start has at most Ch_k^3 choices on the torus. If W_j bounds the number of witnesses rooted at a prescribed site, then

$$W_0 = 1, \quad W_j \leq Ch_j^3 W_{j-1}^2, \quad 2^{-j} \log W_j \leq \sum_{k \geq 1} 2^{-k} \log(Ch_k^3) < \infty.$$

Thus $W_j \leq C_a^{2^j}$. This count may ignore separation, but the probability bound is applied only to the separated witnesses just constructed. Equation (3.9) bounds their union by $(C_a C \sqrt{p})^{2^j}$, which is at most $e^{-2 \cdot 2^j}$ for sufficiently large β . For $h_j \leq h < h_{j+1}$, use $\lambda = \log_a 2$ and $2^j \geq (h/(16a))^\lambda$. Bounded h is absorbed by C ; if h exceeds the torus diameter the event is empty. This proves (3.10). $\square$

Lemma 3.4 (Regularity and localized diffusion). *Assume $\mathsf{l}(P)$ and (3.7). For every deterministic time and anchor, failure of regularity has probability at most*

$$\Delta_A = CA^3 p^6 + C \exp(-cA^{\lambda\nu}). \quad (3.11)$$

Call an exposure good at the anchor if its nonregular times have total length at most $\beta/100$. Its probability of not being good is at most $100\Delta_A$. On a good exposure, the one-turn kernel localized by killing outside $B_A(z)$ has every entry at most $C\beta^{-3/2}$, in either direction.

If a deterministic starting slot is a hole almost surely, localization at its site loses at most $Ce^{-c\beta}$ expected mass in any forward or reversed traversal piece of length at most one turn. This last assertion does not require $\mathsf{l}(P)$.

Proof. There are $O(A^3)$ centers in $B_{11A}(z)$, and a fixed M_* -ball has only a bounded number of six-site subsets. Equation (3.8) therefore gives $CA^3 p^6$ for failure of local regularity. For a

remote blocked site at lifted distance $r > 10A$, failure of the other condition costs at most $C \exp[-c(1+r)^\lambda]$, by (3.10). Indeed all blocked sites of the lift lie in its radius- $\lfloor L/4 \rfloor$ box, so coordinate differences between them are at most $L/2$ and equal their wrapped distances. A crossing in a lifted component is a crossing in the torus component. Summing the remote bound over shells, each containing at most $C(1+r)^2$ sites, gives the second term of (3.11), with a smaller exponential rate. Fubini and Markov give the asserted probability for a good exposure.

Let $v(u)$ be a localized evolving nonnegative mass, initially of mass at most one, and put $H(u) = \|v(u)\|_2^2$. On smooth intervals, with zero extension across the killing boundary,

$$H' = -2\mathcal{D}_t(v).$$

The energy is the sum of $\frac{1}{2}(v_i - v_j)^2$ over unordered real-hole edges, including edges from surviving to killed positions. At a regular time, Lemma 3.2 gives $H' \leq -cH^{5/3}$. At other times and at instantaneous factors, the squared norm cannot increase. Unless it has become zero, $(H^{-2/3})' \geq c'$ throughout the regular times. Each half turn has at least $\beta/2 - \beta/100$ regular time, so its output has 2-norm at most $C\beta^{-3/4}$. The same proof applies to its adjoint, since heat factors are symmetric, transports reverse by bijections, and seam completions reverse by uniform inverse completions. Split a full turn at its temporal midpoint and apply Cauchy–Schwarz to the two resulting masses. This proves the entry bound.

For localization, sample the complete picture under $\mathbb{P}_P$, and independently supply fresh Poisson clocks for the hole walk, accepted where both endpoints are holes. Forced transports use actual picture marks; seam and switch operations do not change sites. An exit from $B_A(z)$ needs at least A inter-site steps. In a traversal piece of length at most β , the physical times run injectively through at most one period, whether forward, backward, or across the seam. Thus the actual marks used by these steps are distinct, as are the fresh marks. Ignore slot compatibility and count site paths. There are boundedly many choices of the next site, slot edge, and actual-or-fresh clock at each step. The factorial bound of Lemma 2.1, combined with the independent fresh factorial measure, bounds the expected number of ordered lists of j steps from the starting site by

$$\frac{(C\beta)^j}{j!}.$$

The denominator is the volume factor for the ordered time simplex. Taking $j = A$ and C_{loc} sufficiently large makes this at most $Ce^{-c\beta}$, using $j! \geq (j/e)^j$. This is an unconditional estimate under the correct exposure marginal; no fresh-clock estimate under the all-hit conditioning is needed. $\square$

3.4 Cutting off the flows

Localized diffusion controls a direct one-turn return. A return after many turns will instead be controlled by an energy inequality. The following finite-torus form of the sparse-source estimate isolates the only error introduced by replacing a pinned boundary with ghost killing.

Lemma 3.5 (Sparse-source pairing on the torus). *At a fixed slice let $(z_i)_{i=1}^{m_T}$ be a sparse list, including repetitions. Retain any subset of its members at which the lifted graph is regular with radius A . For each retained member let $\mu_i \geq 0$ be a mass at most one, supported on the real holes over $B_A(z_i)$. If L satisfies (3.7) and is sufficiently large compared with A , then every real vector Y' on the holes of that slice satisfies*

$$\left| \left\langle \sum_i \mu_i, Y' \right\rangle \right| \leq (Cm_TA^\alpha)^{1/2} \sqrt{\mathcal{D}_t(Y')} + Cm_T L^{1/2-k} \|Y'\|_2. \quad (3.12)$$

Proof. For an individual source slot a in the support of μ_i , take the flow of Lemma 3.2 in the lift about z_i . Multiply its value on each edge by a taper equal to one within anchor radius $L/10$, zero beyond radius $L/5$, and with Lipschitz constant C/L . For instance use a piecewise linear function of the supremum distance of the edge midpoint. The support is strictly inside the lifted box, so this cut flow projects to a flow $\tilde{F}_a$ on the torus real-hole graph.

Its divergence has the form $\mathbf{1}_{\{a\}} + \varepsilon_a$. To see the error bound, compare the taper at an incident edge with its value at the vertex. Their difference is at most C/L ; the untapered divergence is zero except at a , where the taper is one. In the transition annulus every edge is at distance comparable with L from the source, because $A \ll L$. The bounded degree and (3.6) give

$$|\varepsilon_a(j)| \leq CL^{-k-1}, \quad \|\varepsilon_a\|_2 \leq CL^{1/2-k}.$$

No discrepancy arises outside that annulus enlarged by one lattice step, and there are at most CL^3 slots in it.

The projected flow retains the bound $C(1 + d_L(e, x_a))^{-k}$. For two source sites at wrapped distance h , the sum of the product of these envelopes is at most

$$C(1 + h)^{-(2k-3)}. \quad (3.13)$$

For $h \geq 3$, inside the radius- $h/3$ ball about either source, one envelope is at most Ch^{-k} and the sum of the other is at most Ch^{3-k} . Outside both balls, dyadic shells of radii at least $h/3$ contribute at most $C \sum_{r \geq 0} (2^r h)^{3-2k}$. Shell counts on the torus are bounded by the corresponding lattice counts. Since $2k > 3$, this proves (3.13); bounded h follows from square summability of the envelopes.

Sum the cut flows with the source masses as coefficients and denote the result by F . For centers at distance at most $4A$, use the constant overlap bound; for more distant centers, every pair of their source sites is at least half as far apart. With $\sigma = 2k - 3 = 49/50 > \alpha$, sparseness and dyadic annuli give

$$\begin{aligned} \|F\|_2^2 &\leq C \sum_i \left(\#\{j : d_L(z_i, z_j) \leq 4A\} + \sum_{j: d_L(z_i, z_j) > 4A} (1 + d_L(z_i, z_j))^{-\sigma} \right) \\ &\leq Cm_T \left(A^\alpha + \sum_{r \geq 0} (2^r A)^{\alpha-\sigma} \right) \leq Cm_T A^\alpha. \end{aligned}$$

This calculation uses only that each mass is at most one; it permits both overlapping supports and repeated centers. The summed divergence error ε has $\|\varepsilon\|_2 \leq Cm_T L^{1/2-k}$. Finally,

$$\left\langle \sum_i \mu_i, Y' \right\rangle = \sum_e F(e)(Y'_{e-} - Y'_{e+}) - \langle \varepsilon, Y' \rangle.$$

Cauchy-Schwarz, with $\sum_e (Y'_{e-} - Y'_{e+})^2 = 2\mathcal{D}_t(Y')$, proves the claim. The empty retained subset gives zero and is included. $\square$

3.5 Return energy and completion of the induction

We now prove Proposition 3.1. Fix a grid and argue by downward induction over its finite set of real slot points. When every such point is pinned, $\mathfrak{l}(E)$ is immediate. Assume the assertion

for every strict superset of E , and choose a nonempty sparse grid test T disjoint from E . Then $P = E \cup T$ satisfies $\mathbf{l}(P)$, so all the preceding blocking and diffusion estimates hold under $\mathbb{P}_P$.

At an exposure under that law put $m_T = |T|$ and $\zeta = \mathbf{1}_T$. Let $F = 1$ on T , and elsewhere let F be the probability that the ghost-killed walk with kernel U ever hits T at a later cut. Explicitly, start with $F_0 = \zeta$ and iterate $F_{n+1} = 1$ on T , $F_{n+1} = U^*F_n$ off T . These vectors increase to F , with $0 \leq F \leq 1$ and $U^*F = F$ off T . Set $g = F - \zeta$, so $0 \leq g \leq 1$ and $g = 0$ on T .

Let $r(E, T)$ be the mean, over $\mathbb{P}_P$ and the uniform starting point in T , of the indicator that its next pin is in T . The pin inequality of Lemma 2.1 is

$$\mathbb{P}_E(T \subseteq \mathcal{B}_E) \leq (2r(E, T))^{m_T}. \quad (3.14)$$

Conditionally on the exposure, deletion of the other real-pin stops bounds the sum of these next-pin probabilities by $\sum_{i \in T} (U^*F)_i$. Harmonicity off T gives

$$\begin{aligned} \sum_{i \in T} (U^*F)_i &= m_T - \langle F, (I - Q)F \rangle \\ &= \langle \zeta, Q\zeta \rangle + 2\langle Q\zeta, g \rangle - E_g. \end{aligned} \quad (3.15)$$

Indeed the quadratic forms of U, U^*, Q agree on real vectors, and $\langle \zeta, g \rangle = 0$. In particular no reversibility of U has been assumed.

The first term is controlled by one-turn diffusion. Start a unit mass at each $i \in T$ and localize at its site, in either direction. Lemma 3.4 bounds discarded mass in expectation by $Ce^{-c\beta}$. On a good exposure there are at most $C_s(1 + A)^\alpha$ target points of T in its localized region, each receiving at most $C\beta^{-3/2}$. A nongood exposure has probability at most $100\Delta_A$ and still carries total mass at most one. Thus

$$\mathbb{E}_P \langle \zeta, Q\zeta \rangle \leq C m_T (e^{-c\beta} + \Delta_A + A^\alpha \beta^{-3/2}). \quad (3.16)$$

To absorb the linear term in (3.15), split the traversal in direction $d \in \{+, -\}$ at elapsed time $u \in I = [\beta/3, 2\beta/3]$:

$$U_d = B_d(u)C_d(u), \quad Y_d(u) = B_d(u)^*g.$$

The factors use matching hole layers at physical time $t_d(u) = \theta + u$ or $\theta - u$ modulo β , where θ is the cut of T . Each instantaneous factor is assigned to exactly one side; the finitely many exceptional split times do not affect the integrals. As u decreases, $Y_d(u)$ accumulates adjoint factors from the endpoint. Its continuous norm loss is twice its Dirichlet energy, and instantaneous factors contract. Consequently

$$\int_I \mathcal{D}_{t_d(u)}(Y_d(u)) \, du \leq \frac{1}{2} (\|g\|_2^2 - \|U_d^*g\|_2^2) \leq E_g. \quad (3.17)$$

The second inequality follows from the exact identity

$$E_g - \frac{1}{2} (\|g\|_2^2 - \|U_d^*g\|_2^2) = \frac{1}{2} \|g - U_d^*g\|_2^2,$$

because $\langle g, U_d^*g \rangle = \langle g, Qg \rangle$. Also $\|Y_d(u)\|_\infty \leq 1$ and $\|Y_d(u)\|_2 \leq \|g\|_2$.

For each $i \in T$, propagate its unit mass by the starting portion $C_d(u)$, with localization at $z_i = x_i$, and retain it only when the slice is regular at z_i . Call the resulting sum $M_d(u)$, and put

$$J_d = |I|^{-1} \int_I \langle M_d(u), Y_d(u) \rangle \, du.$$

The original pairing $\langle C_d(u)\zeta, Y_d(u) \rangle = \langle U_d\zeta, g \rangle$ is constant in u . Supremum contraction, localization, and the deterministic-time bound (3.11) show that

$$\mathbb{E}_P |\langle U_d\zeta, g \rangle - J_d| \leq C m_T (e^{-c\beta} + \Delta_A). \quad (3.18)$$

This uses Fubini and only regularity at the split, without requiring a good exposure over the entire turn.

Apply Lemma 3.5 to the retained masses. Cauchy–Schwarz in u , (3.17), and $E_g \geq q\|g\|_2^2$ give the pointwise bound

$$|J_d| \leq C \left((m_T A^\alpha / \beta)^{1/2} + m_T L^{1/2-k} / \sqrt{q} \right) \sqrt{E_g}. \quad (3.19)$$

Since $2\langle Q\zeta, g \rangle = \langle U_+\zeta, g \rangle + \langle U_-\zeta, g \rangle$, we complete the square in $\sqrt{E_g}$ to obtain

$$J_+ + J_- - E_g \leq C \left(m_T A^\alpha / \beta + m_T^2 L^{1-2k} / q \right).$$

Combining this with (3.16) and (3.18), and dividing by m_T , yields

$$r(E, T) \leq C \left(e^{-c\beta} + \Delta_A + A^\alpha \beta^{-3/2} + A^\alpha / \beta + m_T L^{1-2k} / q \right). \quad (3.20)$$

Every constant so far is independent of the grid, its number of points, and the induction pin set. Sparseness gives $m_T \leq CL^\alpha$; because $q \geq L^{-5/2}$, the last term of (3.20) is at most

$$CL^{\alpha+7/2-2k} = CL^{-47/100}.$$

The remaining terms, with $A = \lceil C_{\text{loc}}\beta \rceil$, are $o(p)$ as $\beta \rightarrow \infty$: their polynomial orders are $\beta^{-12/5}$, $\beta^{-149/100}$, and $\beta^{-99/100}$, besides stretched-exponential terms. Choose β_0 sufficiently large first. For each fixed $\beta \geq \beta_0$, choose $L_*(\beta)$ sufficiently large for (3.7) and for the last term to make $2r(E, T) \leq p$. Equation (3.14) then proves $l(E)$, completing the finite induction.

For arbitrary finite extra pins and a test at any deterministic time, choose a grid containing their times and zero, and refine it to mesh at most p^{12} . Uniformity in the grid gives the proposition without a limiting procedure. In particular the blocking, component-tail, and regularity conclusions above now hold under every allowed pin law. All time integrals are measurable: the finite-volume histories have finitely many jumps, their kernels are products of measurable heat and instantaneous matrices, and the hitting function is an increasing limit of finite-horizon probabilities. No choice of the auxiliary flows has to be measurable, since only their deterministic pairing inequality is used.

Corollary 3.6 (Positive spontaneous magnetization). *For every fixed S and all sufficiently large finite β ,*

$$m_{S,\beta} \geq S(1 - \beta^{-9/10}) > 0.$$

Proof. The passage from pinned-cycle probabilities to the pressure slope uses pressure increments, as in [14, Section 7]. Fix β above the threshold of Proposition 3.1, enlarged so that $\beta > 1$, and put $p = \beta^{-9/10}$. For an exponent field $s > 0$, set $q(s) = 1 - e^{-s}$, and use the ghost picture with no extra real pins. A real slot is down with probability one half of its probability of belonging to a cycle missing the ghost pins: pinned cycles are up, and missing cycles are colored fairly. The singleton case of Proposition 3.1 therefore gives a down probability at most $p/2$, whenever $L \geq L_*(\beta)$ and $q(s) \geq L^{-5/2}$.

Define the finite-volume exponent pressure

$$\mathcal{P}_L(s) = V^{-1} \log \text{Tr} \exp \left(H_0 + s \sum_x Y_x \right).$$

Write $\langle \cdot \rangle_s$ for the corresponding tilted Gibbs expectation. There are ℓV slots and $\ell = 2S$, so differentiation and the color representation give

$$\mathcal{P}'_L(s) = V^{-1} \left\langle \sum_x Y_x \right\rangle_s \geq S(1-p).$$

For fixed $0 < a < b$, this holds simultaneously for all $s \in [a, b]$ once $L \geq \max\{L_*(\beta), q(a)^{-2/5}\}$. Integration yields

$$\mathcal{P}_L(b) - \mathcal{P}_L(a) \geq S(1-p)(b-a).$$

By rotation invariance and the pressure limit (1.2), $\mathcal{P}_L(s) \rightarrow \mathcal{P}(s) = \beta p_{S,\beta}(s/\beta)$ along even L . Pass to this limit first. The limiting pressure is S -Lipschitz, because $\|\sum_x Y_x\| \leq SV$; letting $a \downarrow 0$ therefore gives $\mathcal{P}(b) - \mathcal{P}(0) \geq S(1-p)b$. Finally divide by b and let $b \downarrow 0$. Since $\mathcal{P}'(0+) = p'_{S,\beta}(0+) = m_{S,\beta}$, this proves the claim with the volume limit taken before removal of the field. $\square$

4 Spatial comparison at a vanishing field

The sparse-pin estimate controls the missing cycles at prescribed points. We now turn it into a bound for a spatially oscillating magnetization. The two steps are quite different. First, an exact susceptibility identity expresses the transverse square through a one-particle resolvent on the holes. Second, we show that the resolvent is nearly constant on large rectangular cells. Small closed classes of holes are allowed; a trace identity controls their total contribution.

Proposition 4.1 (Weak-field spatial estimate). *Fix S . For all sufficiently large β there are $C_\beta, R_0(\beta) < \infty$ with the following property. Let L be even, let $R_0(\beta) \leq R \leq L$, and partition Λ_L into rectangular cells whose side lengths belong to $[R/2, 4R]$. For every deterministic real $f : \Lambda_L \rightarrow [-1, 1]$ with $\sum_{x \in B} f_x = 0$ on each cell B , put*

$$D_f = \sum_x f_x Z_x, \quad q = R^{-5/2}, \quad w = 1 - q = e^{-b}.$$

If $\langle \cdot \rangle_b$ is the Gibbs expectation with exponent $H_0 + b \sum_x Y_x$, then

$$\frac{q}{V} \langle D_f^2 \rangle_b \leq C_\beta R^{-1/40}. \quad (4.1)$$

The constants are uniform in the partition, f , and L .

Throughout the proof β is fixed above a sufficiently large threshold; subsequent lower bounds on R may depend on β . We use the soft-pin representation of Section 3, with no extra deterministic real pins. Expose all cycles missing the soft pins. Write $\mathbf{a}$ for their trajectories, A_0 for their real slots at the cut, and $\mathcal{H}$ for the complementary cut slots, called the holes. The cut is after the seam and its soft-pin insertions, which do not move any line. The soft pins themselves may be placed on real cut slots, instead of using the equivalent ghosts. A hole cycle is then pinned and has up color; every exposed cycle receives an independent fair color.

4.1 The susceptibility as a hole resolvent

Condition on $\mathbf{a}$. The fresh real-hole motion consists of rate-1/2 jumps across the currently available slot edges, the forced hole transports at exposed jumps, and a uniform completion of the seam permutation within each site. Its one-turn kernel, without killing or conditioning on its cycles, is denoted by K . In this section we use the *row-forward convention*: K_{ij} is the probability of going from i to j . Thus K is doubly stochastic. Set

$$D = I - wK, \quad D^{-1} = \sum_{r \geq 0} w^r K^r. \quad (4.2)$$

This inverse exists and is entrywise nonnegative. Empty hole sets use empty matrices, determinant one, and sums zero.

Lemma 4.2 (Soft-pin identities). *Given the exposure $\mathbf{a}$, for $i, j \in \mathcal{H}$, $i \neq j$,*

$$\mathbb{P}(i \text{ is pinned} \mid \mathbf{a}) = q(D^{-1})_{ii}, \quad (4.3)$$

$$\mathbb{P}(i, j \text{ are pinned, next pin from } i \text{ is } j \mid \mathbf{a}) = \frac{q^2}{w}(D^{-1})_{ij}. \quad (4.4)$$

Moreover,

$$\mathbb{E} \operatorname{Tr} D^{-1} \leq N. \quad (4.5)$$

For $f_i = f_{x_i}$ on the slots,

$$\langle D_f^2 \rangle_b = \frac{1}{4} \sum_i f_i^2 + \frac{1}{2} \mathbb{E} \sum_{\substack{i, j \in \mathcal{H} \\ i \neq j}} f_i f_j (D^{-1})_{ij}. \quad (4.6)$$

All expectations here include the normalized picture and soft-pin law.

Proof. In the exposure disintegration, the proposed exposed paths have a fixed cycle weight and a factor $w^{|A_0|}$, since none of their cut slots is pinned. For a proposed pin set $P' \subseteq \mathcal{H}$, put $Q = \mathcal{H} \setminus P'$. Lemma 2.2, with the row convention obtained by transposition, says that the remaining cycles all meet P' with probability $\det(I - K_{QQ})$. Consequently the unnormalized conditional weight of P' is

$$q^{|P'|} w^{|\mathcal{H}| - |P'|} \det(I - K_{QQ}). \quad (4.7)$$

Multilinearity of the determinant in the rows gives

$$\sum_{P' \subseteq \mathcal{H}} q^{|P'|} w^{|\mathcal{H}| - |P'|} \det(I - K_{QQ}) = \det(I - wK).$$

Restricting this sum to sets containing i gives q times the diagonal cofactor of D , proving (4.3).

For (4.4), keep i, j pinned. At fixed Q the all-hit probability times the conditional next-pin entry is

$$\det \begin{pmatrix} K_{ij} & K_{iQ} \\ -K_{Qj} & I - K_{QQ} \end{pmatrix}.$$

This is a polynomial identity, including when $I - K_{QQ}$ is singular, by the unnormalized identity in Lemma 2.2. Put $G = \mathcal{H} \setminus \{i, j\}$. Summing over the Bernoulli choices on G replaces this expression by

$$\det D_{GG} (K_{ij} + w K_{iG} D_{GG}^{-1} K_{Gj}).$$

Indeed an unselected row in the bordered determinant is an identity row with zero border entry, and a selected row has probability w . The principal submatrix D_{GG} is invertible because $w < 1$. The two-by-two Schur complement of D_{GG} in D now gives the last display as $\det D (D^{-1})_{ij}/w$. Multiply by q^2 and divide by the normalizing determinant to obtain the assertion.

Condition instead on the picture and its colors. Each up cut slot is independently pinned with probability q , and no down slot can be pinned: summing its two allowed Bernoulli weights gives $w + q = 1$ for up, and only w for down. The expected total number of pins is therefore at most qN . Pins never lie in A_0 , so summing (4.3) and taking expectations proves (4.5).

For the transverse identity rotate coordinates so that the field component is s^3 and the tested component is s^1 . The normalized field matrix is $\text{diag}(1, w)$. Split it symmetrically around each insertion; this is permitted because the total field commutes with H_0 . At a transverse insertion its slot matrix is

$$\text{diag}(1, \sqrt{w}) s^1 \text{diag}(1, \sqrt{w}) = \frac{\sqrt{w}}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For distinct i, j , expand pins on all other slots. Two flips give zero unless they lie on the same cycle. On that cycle one of the two arcs between the flips must be down and must contain no other pin. Adding pins at i, j therefore identifies each allowed coloring with one of the events “next pin from i is j ” or “next pin from j is i .” If there are no other pins on this cycle both events contribute, exactly accounting for its two allowed colorings. The ratio of insertion weights to the weights with the added pins is $w/(4q^2)$. Formula (4.4) thus gives $\frac{1}{4}\mathbb{E}[(D^{-1})_{ij} + (D^{-1})_{ji}]$ for the two-slot expectation. Sum against $f_i f_j$ over ordered distinct pairs. Here inverse entries are interpreted as zero if either cut slot is outside the random set $\mathcal{H}$. The same-slot identity $(s^1)^2 = I/4$ gives the first term in (4.6). The seam projection ensures that these slot sums are precisely the physical site-spin observable. $\square$

Define the bounded real vector

$$u = qD^{-1}f_{\mathcal{H}}, \quad e_i = \sum_{j \in \mathcal{H}} K_{ij}(u_i - u_j)^2. \quad (4.8)$$

Since qD^{-1} is stochastic, $|u_i| \leq 1$. Double stochasticity and $Du = qf_{\mathcal{H}}$ imply

$$\frac{1}{2} \sum_i e_i = \langle u, (I - K)u \rangle = \frac{q}{w} (\langle u, f_{\mathcal{H}} \rangle - \|u\|_2^2) \leq CqN. \quad (4.9)$$

No reversibility of K is used here.

We will approximate u by a constant on the holes in each cell. The missing slots mean that $f_{\mathcal{H}}$ no longer has exact cellwise cancellation. The following estimate supplies its replacement.

Lemma 4.3 (Cancellation after exposure). *For every cell B ,*

$$\mathbb{E} \left| \sum_{\substack{i \in \mathcal{H} \\ x_i \in B}} f_i \right| \leq C\sqrt{|B|/q}. \quad (4.10)$$

Proof. Return to the original axes, with field direction Y , and put $T_B = \sum_{x \in B} f_x Y_x$. The ordinary Laplace transform $F_B(z) = \langle e^{zT_B} \rangle_b$ is zero-free on $|z| < cb$: all slot fields in direction Y have

positive real part there, so Lemma 2.3 applies. Take its analytic logarithm equal to zero at 0. Spectral calculus for T_B gives

$$\operatorname{Re} \log F_B(z) \leq |z|S|B| \leq Cb|B| \quad (|z| < cb).$$

The Borel–Carathéodory inequality on this disk, followed by Cauchy’s second-derivative estimate on a smaller concentric disk, therefore yields

$$\operatorname{Var}_b(T_B) = (\log F_B)''(0) \leq C|B|/b \leq C|B|/q.$$

Translation invariance and $\sum_{x \in B} f_x = 0$ give $\langle T_B \rangle_b = 0$. In the color representation its conditional expectation given $\mathbf{a}$ is

$$\mathbb{E}(T_B \mid \mathbf{a}) = -\frac{1}{2} \sum_{\substack{i \in A_0 \\ x_i \in B}} f_i = \frac{1}{2} \sum_{\substack{i \in \mathcal{H} \\ x_i \in B}} f_i.$$

Here exposed cycles are fair and every hole cycle is up; the total sum of the slot coefficients on B is zero. Conditional Jensen and Cauchy–Schwarz prove (4.10). $\square$

4.2 A graph recording communication between holes

The small traversal energy in (4.9) must now control spatial variation. A hole can be carried by the exposed trajectories even when the fresh walk makes no jumps. We retain that motion explicitly in a graph on the cut holes.

Ignore fresh walk jumps and follow only forced transports from the cut to the seam. Choose one allowed residual seam matching within each site. The resulting lines, called *skeleton lines*, give a permutation ϕ of $\mathcal{H}$. Define an undirected graph Γ on $\mathcal{H}$ with the following edges:

- (i) an endpoint edge from i to every possible final cut hole under a residual seam matching, including $\phi(i)$;
- (ii) a contact edge between i, j whenever their skeleton lines occupy adjacent sites, with an available slot edge between them, throughout some interval of positive length.

Self-edges may be omitted. All choices of representatives and matching can be made by fixed orderings of the finite sets.

Lemma 4.4 (Communication costs and closed classes). *An endpoint edge from i to j satisfies $(u_i - u_j)^2 \leq C_\beta e_i$. For a contact edge witnessed by a fixed-position interval of length $\tau \leq 1$,*

$$(u_i - u_j)^2 \leq C_\beta \tau^{-1} (e_i + e_j). \quad (4.11)$$

The kernel K is closed on every connected component of Γ , and

$$\mathbb{E} \#\{\text{components of } \Gamma\} \leq qN. \quad (4.12)$$

Proof. The walk jump rate is uniformly bounded by a constant depending on ℓ . The probability of making no fresh jump during a period is at least e^{-C_β} ; the probability of any prescribed allowed seam image is at least $1/\ell$. Thus an endpoint entry of K is at least c_β . For a contact interval, require exactly one jump from line i to line j during that interval and no other jumps, and choose endpoint $\phi(j)$. This gives $K_{i, \phi(j)} \geq c_\beta \tau$. Compare u_i to u_j through $u_{\phi(j)}$, using also the endpoint bound for j ; this proves (4.11).

A fresh walk changes skeleton line only through contact, and every possible seam endpoint has been included in Γ . Hence each component is closed under K . Its stochastic block has eigenvalue 1. Every eigenvalue λ of that block satisfies $|\lambda| \leq 1$, and

$$\operatorname{Re} \frac{1}{1 - w\lambda} = \frac{1 - w \operatorname{Re} \lambda}{|1 - w\lambda|^2} > 0.$$

Its contribution to $\operatorname{Tr} D^{-1}$ is therefore at least $1/q$, from the eigenvalue 1. This trace argument applies equally to non-diagonalizable blocks. Sum over the components and use (4.5). $\square$

Every edge of Γ also has a spatial realization. An endpoint edge follows one skeleton line and then the seam, which does not change its site. A contact edge follows its first skeleton line to a contact time, crosses one site edge, and follows the second line back to its cut. Such a realization consists of at most two pieces of actual marks in chronological or reverse chronological order, and one extra site edge. This observation will prevent an edge leaving a small obstacle from making a long excursion or winding around the torus.

4.3 Coarse obstacles and local communication

Choose $d = \lceil C_0 \beta \rceil$, with the fixed constant C_0 large enough below. Partition each coordinate circle into integer intervals of lengths in $[d, 2d]$. Their product boxes are indexed by a three-dimensional coordinate torus. Fine distances refer to site coordinates and coarse distances to these box indices; both use the supremum norm and the periodic convention. A site is *blocked* if it has no cut hole.

We use the following deterministic input from [15, Lemma 5.5], also recalled in [14, Section 3]. For each fixed a there is $\rho(3, a) \geq a + 1$ such that two unblocked sites at distance at most a can be joined by a nearest-neighbor unblocked path in the ball of radius $\rho(3, a)$ about the first, of length at most $(2\rho(3, a) + 1)^3$, provided that ball contains fewer than six blocked sites. We use $a = 44$ and fix $M \geq \max\{48, \rho(3, 44) + 2\}$. The result concerns arbitrary blocked sets; we apply it in local lifts to the lattice. Increase C_0 , if necessary, so that d dominates these fixed geometric constants.

Let $0 < \delta < \min\{1, \beta/2\}$; its value in terms of R will be chosen only after the comparison is proved. Put $j_* = \lfloor d/20 \rfloor$. Declare a coarse box *bad* when at least one of the following occurs:

- (i) some M -ball centered within fine distance $10d$ of the box contains at least six blocked sites;
- (ii) starting within that $10d$ -neighborhood, a site path of j_* inter-site steps is supplied by distinct actual marks in chronological or reverse chronological order during one period;
- (iii) among marks on edges incident to the $15d$ -neighborhood, there is a mark within δ of 0 or β , or two distinct marks at times less than δ apart.

The last event will be called a *short-time flag*. Slot compatibility is deliberately not required in (ii); this makes the condition constrain every skeleton trajectory. Join bad boxes at coarse distance at most eight. All probabilities below use the original joint picture law. The bad events may depend on the full picture, even though K and u depend only on its exposure.

Lemma 4.5 (Bad-box tails). *There are $\beta_0, c, C, \lambda_1 > 0$ and, for each $\beta \geq \beta_0$, a number $\delta_0(\beta) > 0$ such that for every $0 < \delta \leq \delta_0(\beta)$, uniformly in the admissible volumes and pin parameters of this section,*

$$\mathbb{P}\{z \text{ is bad and its bad component reaches distance } h \text{ from } z\} \leq C e^{-ch^{\lambda_1}}. \quad (4.13)$$

A specified box has a short-time flag with probability at most $C_\beta \delta$.

Proof. Factorial domination from Lemma 2.1 bounds the one-mark count near the ends of the period and the ordered two-mark count with time gap less than δ by $C_\beta \delta$. For a prescribed starting site, the expected number of ordered paths using j_* marks is at most $(C\beta)^{j_*}/j_*$: the bounded choices of a next slot edge and the factorial factor are absorbed in C , and the ordered time simplex has volume β^{j_*}/j_* . Summing starting sites gives the bound

$$\varepsilon_{\text{mark}} = Cd^3(C\beta)^{j_*}/j_*! + C_\beta \delta \quad (4.14)$$

for the union of (ii) and (iii). Such paths stay within fine distance $11d$ of the box, so all their data are in its fixed $15d$ -neighborhood, with a harmless one-edge enlargement. Take C_0 large; the first term is then at most $Ce^{-c\beta}$.

Here is the dependence estimate needed to pass from single boxes to components. Choose a large integer a with $a^{1/100} > 4$, and a fixed coarse radius h_0 large enough that data neighborhoods at separated leaves below are disjoint. Set $h_k = h_0 a^k$. A bad path from a prescribed root to distance h_k contains two crossings at scale h_{k-1} : one starts at the root, and one at its first visit to distance $h_k/2$. Truncate each on its first reach of the smaller radius. Their ranges are separated by at least $h_k/4$, since

$$h_k/2 - 2(h_{k-1} + 8) \geq h_k/4.$$

Recursion gives 2^k distinct bad leaves with the sparse growth bound of Proposition 3.1, in coarse coordinates. Indeed a ball of radius less than $h_r/8$ meets at most one branch at every split of scale at least h_r , and hence at most 2^{r-1} leaves; the choice of a gives the growth exponent $1/100$ also between consecutive scales. The number of possible witnesses is at most $C_1^{2^k}$, because at a split of scale h_r there are at most Ch_r^3 choices and $\sum_{r \geq 1} 2^{-r} \log(Ch_r^3) < \infty$.

For any specified subset of leaves of type (i), choose a witnessing center and six blocked sites. There are at most Cd^3 choices per leaf, since M is fixed. Select one fixed slot at each chosen site. These fine-slot tests are sparse with a constant independent of d : sites in a fine ball of radius r come from leaves in a coarse ball of radius $C(1 + r/d)$, so their number is at most $6C[1 + C(1 + r/d)]^{1/100} \leq C_s(1 + r)^{1/100}$. Leaf separation makes the selected sites distinct. A blocked site has all its slots exposed, so Proposition 3.1 bounds the joint probability for m such leaves by $(Cd^3 p^6)^m$, where $p = \beta^{-9/10}$.

For any subset of mark-type leaves, the data neighborhoods are disjoint. Expand their indicators by the path and flag counts used above and apply factorial domination to all the selected marks together. Distinct neighborhoods use distinct marks; their integrals factor. This gives the product bound $\varepsilon_{\text{mark}}^m$, without assuming independence of the weighted picture. In every witness at least half the leaves have one of these two types. Sum over that subset and over the witnesses. Since $d^3 p^6 = O(\beta^{-12/5})$, choosing β large and then $\delta_0(\beta)$ small puts both leaf costs below a fixed sufficiently small witness budget. The probability is then at most Ce^{-c2^k} , with c, C uniform for $\delta \leq \delta_0(\beta)$. This proves (4.13) with $\lambda_1 = \log_a 2$, after changing constants for intermediate and bounded radii. The argument uses only wrapped distances and the bound $|B_h| \leq C(1 + h)^3$, so it is valid on the coarse torus. $\square$

Lemma 4.6 (Local paths near good boxes). *Choose a representative cut hole $i(z)$ in every good box z . Any two cut holes in one good box, and representative holes in two nearest-neighbor good boxes, can be joined in Γ using at most C_β contact edges, all with contact duration at least δ , whose cut starts stay within fine distance C_β of the boxes.*

Moreover, if a spatial realization of any edge of Γ meets a good box z , its whole realization lies within fine distance $d/2$ of a point in that box, after increasing the fixed lower threshold on d . In particular both cut endpoint boxes have coarse distance at most one from z .

Proof. A good box contains an unblocked site, since otherwise an M -ball in it would contain six blocked sites. To connect two desired sites, first use a coordinate path of length Cd through the one- or two-box neighborhood. Replace every blocked vertex of this path by a surviving nearest neighbor. Such a neighbor exists: six blocked neighbors would contradict (i). Consecutive replaced vertices have distance at most three. The imported local-connectivity result joins them through unblocked sites with bounded length and displacement. All its balls lie in the $10d$ -neighborhood where (i) is absent. Lift the resulting path to cut slots, using the complete adjacency of available slots across a site edge. If different slots at one site need joining, pass through a surviving neighbor of that site. The absence of (iii) means that these cut adjacencies persist for at least δ after the cut. This proves the first assertion.

For the second assertion, a realization has at most two monotone mark pieces and a contact edge. Suppose one piece meets z . The portions on either side of that visit each have fewer than j_* steps: a longer portion, read in the appropriate time direction, would violate (ii). If there is a second piece, its contact endpoint is therefore within $2j_* + 1$ of the visit. This point lies in the tested $10d$ -neighborhood, and (ii) bounds the whole second piece by $j_* - 1$ steps. If the contact edge itself meets z , apply the same argument from its endpoints. Thus every point of the realization lies within $3j_* + 2 < d/2$ of the visit. This reasoning also applies in a lift: every monotone piece uses distinct torus marks during one period. $\square$

The first assertion converts ordinary paths through good boxes into energy bounds. The second is the essential control on a hole trapped among bad boxes: any escape through a good boundary must take place locally. We next build those boundaries on a finite portion of the covering lattice, including when the cell under consideration is the whole torus.

4.4 Filled hulls and the comparison function

We recall precisely the deterministic hull construction that we import from [15, Lemma 6.1 and Section 6.1, Equations (6.2)–(6.3)]. For any finite blocked set $\mathcal{Z} \subset \mathbb{Z}^3$, join its points at supremum distance at most eight. For each component C , thicken it to $A_C = \{z : \text{dist}(z, C) \leq 4\}$ and fill the finite nearest-neighbor components of $\mathbb{Z}^3 \setminus A_C$. The resulting hull H_C is contained in the coordinate bounding box of A_C . The hulls are disjoint or nested, so the maximal ones are disjoint. Their exterior vertex boundaries consist of unblocked vertices and are connected by unblocked nearest-neighbor paths in the unit enlargement of that bounding box. At an edge from H_C to its complement the exterior vertex has distance five from C and at least four from every other blocked component.

Choose an exterior-boundary representative r_H for each maximal hull and define $\rho(z) = r_H$ on H , fixing other vertices. Apply this map *once*. Each lattice edge has a route between its two images, using only unblocked vertices and lying in the original edge together with the enlarged boxes of its incident hulls. For completeness, if its endpoints belong to distinct maximal hulls H, H' , the route is

$$r_H \longrightarrow v \longrightarrow u \longrightarrow r_{H'} \quad (u \in H, v \in H').$$

The first and last portions run along the respective exterior boundaries, and the middle portion reverses the original edge. A representative can lie in another hull, so ρ is not assumed idempotent. This is why the once-applied convention matters. These inputs concern arbitrary finite blocked sets and require no regular-anchor hypothesis.

Let $n \geq 1$ be an integer parameter and assume $10^6 dn \leq R \leq L$. For a cell B , lift its coordinate intervals to $\mathbb{Z}^3$, and take a coarse rectangular window containing all boxes meeting that lift, padded

by $1000n$ coarse units. Copy the torus box data periodically into the window and declare precisely its bad boxes blocked; outside the window there are no blocked boxes. Call B *usable* if every component of this finite blocked set has diameter at most $2n$. Lemma 4.5 gives

$$\mathbb{P}(B \text{ is unusable}) \leq C_\beta R^3 e^{-cn^{\lambda_1}}. \quad (4.15)$$

To verify the periodic issue in this bound, a component of diameter greater than $2n$ has, from one of its vertices, a path to distance n . Truncate at its first reach. The path stays within $n + 8$ coarse units of its root; the assumed bound $10^6 dn \leq L$ ensures that its projection preserves these distances and gives a torus bad component reaching n . The window has at most $C_\beta R^3$ vertices, including its periodic copies, which proves the union bound. The hull construction therefore takes place on a finite blocked subset of the covering lattice.

For usable B , apply the preceding lattice hull construction. Each enlarged bounding box has diameter at most $2n + 10$; simple routes in the union of at most two such boxes have length at most Cn^3 . All hulls and routes used near B are strictly inside the padded window. There the finite-set designation “unblocked” means that the corresponding torus coarse box is good. Choose representative cut holes periodically for all good boxes and define on fine sites of the lifted cell

$$U_B(x) = u_{i(\rho(z_x))}, \quad (4.16)$$

where z_x is the coarse box containing x . Thus $|U_B| \leq 1$. This is a function on all sites, even when some sites have no hole.

Lemma 4.7 (Comparison across a cell and out of a hull). *There is a constant C_β such that the following hold for every usable B . For fine nearest neighbors x, x' within its lift,*

$$|U_B(x) - U_B(x')|^2 \leq C_\beta n^6 \delta^{-1} \sum_{\substack{i \in \mathcal{H} \\ \text{dist}(x_i, x) \leq C_\beta n}} e_i. \quad (4.17)$$

For a cut hole j above $x \in B$, the same right side bounds $|u_j - U_B(x)|^2$, unless either

- (i) there is a short-time flag in some box at coarse distance at most $100n$ from z_x ; or
- (ii) the Γ -component of j has at most $C_\beta n^3$ vertices.

The distances in the energy sum may be taken on the torus.

Proof. If x, x' lie in the same coarse box the left side of (4.17) vanishes. Otherwise their box indices are nearest neighbors. The imported hull route joins their images under ρ through good boxes, in a region of diameter Cn , with length at most Cn^3 . Replace each of its coarse edges by the cut-hole paths of Lemma 4.6. This gives at most $C_\beta n^3$ edges of Γ in the stated neighborhood, each obeying (4.11) with $\tau = \delta$. Telescoping and Cauchy–Schwarz bound the total squared difference by path length times the sum of these costs. Bounding the multiplicity of any e_i by the path length itself gives the factor $C_\beta n^6 \delta^{-1}$.

Now let j lie over x . If z_x is outside all maximal hulls, it is good and the local paths already prove the assertion. Otherwise let H be its maximal hull. This hull and its enlarged box have diameter $O(n)$; they project injectively onto the coarse torus by the same bound $10^6 dn \leq L$. The set J_H of cut holes whose boxes belong to its projection has size at most $C_\beta n^3$. If the Γ -component of j stays in J_H , exception (ii) applies. Otherwise take a simple Γ -path from j to its first vertex outside J_H . It has at most $|J_H|$ edges.

To locate these edges, lift a spatial realization starting at its endpoint in H . If the realization stays in H , it is local. If it leaves, its first boundary crossing has an exterior coarse box z with

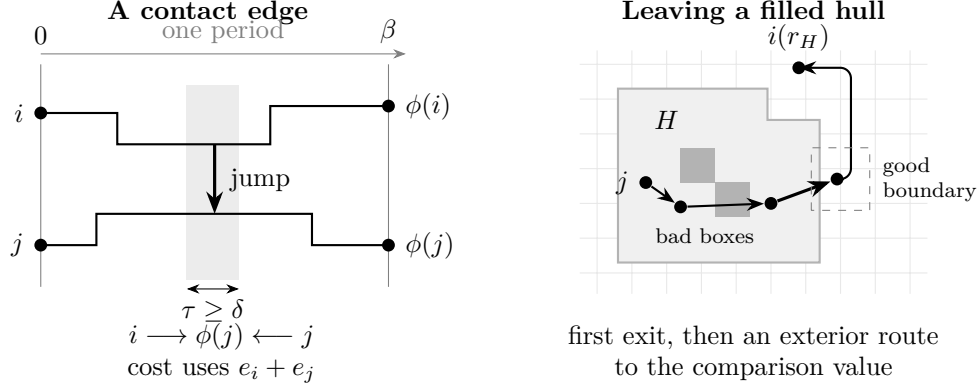


Figure 2: Two parts of the hole comparison, drawn schematically. Left: a fresh walk jumps between skeleton lines during a contact interval. Comparing both cut values through $u_{\phi(j)}$ charges energy only to the cut starts i, j . Right: a path in Γ first exits a filled coarse hull. Clearance makes its crossing box good; the absence of long mark paths then confines the whole exiting link near that box. A surviving exterior-boundary route leads to $i(r_H)$, which defines U_B on the hull. Gray squares indicate bad boxes; the larger gray region represents the filled hull, with distances and thicknesses not to scale.

distance at least four from every bad box. This crossing lies in the interior of the data window, so z is genuinely good on the torus. Lemma 4.6 confines the *entire* realization to the $d/2$ -neighborhood of that crossing, and its cut endpoint boxes to coarse distance one from z . In particular it cannot wind around the torus and return to another copy of H . For every edge on the first-exit path, its realization and both cut starts thus remain within coarse distance Cn of z_x .

It remains to justify its contact costs; the bad boxes inside H did not by themselves guarantee any minimum contact duration. Choose a contact time on its localized realization, and extend to the maximal interval on which the two skeleton positions are fixed. An endpoint of this interval is 0 , β , or an actual mark incident to one of the two occupied slots. If the interval had length less than δ , its endpoints would give either a mark within δ of an end of the period or two distinct incident marks within δ of each other. The contact sites are within $12n$ coarse units of z_x : the hull diameter is at most $2n + 8$, its boundary boxes are within $2n + 9$ of z_x , and a realization meeting such a box stays within one further coarse unit. Thus $2n + 10 \leq 12n$. Both endpoint marks are incident to these same fixed positions, including when an endpoint time precedes the chosen contact time. Such a flag is excluded by (i). Every contact therefore has duration at least δ . Endpoint edges need no duration estimate.

For the exiting edge, take its boundary-crossing exterior box z as above. The final cut hole lies in a box at coarse distance at most one from z . Clearance makes all boxes within that distance good. A coordinate coarse path between them stays in this good neighborhood, and the local hole paths lead to $i(z)$. Finally, z and the representative r_H both lie on the exterior vertex boundary of H . The imported boundary route joins them through good boxes in its enlarged bounding box, using at most Cn^3 steps after loop erasure. Lift this route by the local hole paths. We have joined j to $i(r_H)$, the hole defining $U_B(x)$, by at most $C_\beta n^3$ controlled edges, all using cut starts within fine distance $C_\beta n$ of x . The same telescoping estimate proves the assertion. $\square$

Figure 2 illustrates the two local comparisons used in this argument: energy is charged at cut starts, and an exit from a hull reaches a nearby surviving boundary route.

4.5 Energy, exceptional sets, and the final estimate

The construction has supplied a bounded comparison function on each usable cell. Its lattice gradients are controlled by the total traversal energy, and a hole fails to match that function only near a short-time flag or in a small closed class. We now sum these three contributions and choose the auxiliary scales.

Lemma 4.8 (Cellwise approximation of the resolvent). *There are random numbers $c_B \in [-1, 1]$, one per cell, such that*

$$\mathbb{E} \sum_B \sum_{\substack{i \in \mathcal{H} \\ x_i \in B}} (u_i - c_B)^2 \leq C_\beta N R^{-1/20}. \quad (4.18)$$

Proof. On a usable cell let $c_B = |B|^{-1} \sum_{x \in B} U_B(x)$; on an unusable cell put $c_B = 0$. The rectangular lattice Poincaré inequality, whose constant is at most CR^2 for the stated side lengths, gives

$$\sum_{x \in B} |U_B(x) - c_B|^2 \leq CR^2 \sum_{\substack{\{x, x'\} \subset B \\ |x - x'|_1 = 1}} |U_B(x) - U_B(x')|^2.$$

It holds in the chosen lift, also for a cell covering an entire coordinate circle: only the internal rectangle edges are needed. Use Lemma 4.7 both here and for the nonexceptional differences $u_i - U_B(x_i)$. There are at most ℓ holes above a site. In summing all the local energy sums, any fixed torus e_i occurs at most $C_\beta n^3$ times, since the original fine sites, over all cells, partition the torus. Periodic copies used to construct the windows do not add summation sites. Equation (4.9) therefore bounds these contributions by

$$C_\beta R^2 n^9 \delta^{-1} q N.$$

All remaining squared differences are at most four. A fixed fine site has at most Cn^3 boxes in its short-flag neighborhood; its exceptional probability is at most $C_\beta n^3 \delta$ by Lemma 4.5. The total number of holes in Γ -components of size at most $C_\beta n^3$ is at most $C_\beta n^3$ times the number of components, whose expectation is bounded by (4.12). Finally sum (4.15) with weights at most $\ell|B|$. Together these estimates give

$$\frac{1}{N} \mathbb{E} \sum_B \sum_{\substack{i \in \mathcal{H} \\ x_i \in B}} (u_i - c_B)^2 \leq C_\beta \left(R^2 n^9 \delta^{-1} q + n^3 \delta + n^3 q + R^3 e^{-cn^{\lambda_1}} \right). \quad (4.19)$$

Take now

$$\delta = R^{-1/8}, \quad n = \lceil R^{1/200} \rceil.$$

At fixed β , sufficiently large R meets all preceding smallness conditions, including $10^6 dn \leq R \leq L$, and the short-flag threshold in Lemma 4.5. The first three terms on the right of (4.19) are, respectively,

$$O(R^{-33/100}), \quad O(R^{-11/100}), \quad O(R^{-497/200}).$$

The fourth is smaller than any negative power of R . This proves (4.18) with the stated, nonoptimal exponent. $\square$

Proof of Proposition 4.1. The partition has at most CV/R^3 cells and each has volume comparable to R^3 . Lemmas 4.8 and 4.3, together with $|c_B| \leq 1$, give

$$\begin{aligned} \mathbb{E}|\langle f_{\mathcal{H}}, u \rangle| &\leq \mathbb{E} \sum_B \sum_{i \in \mathcal{H}, x_i \in B} |u_i - c_B| + \sum_B \mathbb{E} \left| \sum_{i \in \mathcal{H}, x_i \in B} f_i \right| \\ &\leq C_\beta N R^{-1/40} + CV(qR^3)^{-1/2}. \end{aligned}$$

In (4.6), add the diagonal terms inside the resolvent sum and discard their nonnegative subtraction. Since $u = qD^{-1}f_{\mathcal{H}}$, this yields

$$\frac{q}{V} \langle D_f^2 \rangle_b \leq \frac{qN}{4V} + \frac{1}{2V} \mathbb{E}|\langle f_{\mathcal{H}}, u \rangle| \leq C_\beta(q + R^{-1/40} + R^{-1/4}).$$

This is (4.1). The assumptions needed for the sparse-pin estimate hold because $q = R^{-5/2} \geq L^{-5/2}$ and the lower bound on R can include $L_*(\beta)$. $\square$

5 Removing the field and comparing spatial scales

Proposition 4.1 controls an ordinary transverse square in a small positive field. We first turn this estimate into an exponential bound at zero field. The bound is weak for one prescribed test, but it holds uniformly over enough tests to compare magnetization averages in a cell with those in its children. Throughout this section, β is sufficiently large and fixed.

5.1 Curvature of a logarithmic Laplace transform

The analytic step uses the following strip lemma. Its positivity is essential: small curvature at the origin will then control curvature on a real interval whose length grows logarithmically with the spatial scale.

Lemma 5.1 (Strip curvature). *Let F be holomorphic on $\{z \in \mathbb{C} : |\operatorname{Im} z| < a_0\}$, real and even on the real axis, with $F(0) = 0$. Suppose that, for some $K < \infty$,*

$$0 \leq F(x) \leq K|x|, \quad \operatorname{Re} F(x + iy) \leq F(x) \quad (x \in \mathbb{R}, |y| < a_0).$$

For each $a \in (0, a_0)$ there is a strictly positive Schwartz function J_a such that, with $g_a(x) = F(x) - \operatorname{Re} F(x + ia)$,

$$F'' = J_a * g_a, \quad J_a = \frac{2}{a^2} (h_a * h_a), \quad h_a(x) = \frac{\pi}{2a} \operatorname{sech}^2\left(\frac{\pi x}{a}\right). \quad (5.1)$$

In particular, with $c_a = 2\pi/a$,

$$0 \leq F''(x) \leq e^{c_a|x|} F''(0), \quad F(x) \leq \frac{F''(0)}{c_a^2} (e^{c_a|x|} - 1 - c_a|x|). \quad (5.2)$$

Proof. Put $u(x, y) = \operatorname{Re} F(x + iy)$. Schwarz reflection makes u even in y . We first justify Fourier transformation in the unbounded horizontal direction. Fix $a' < a_0$. Around each vertical segment $\{x + iy : |y| \leq a'\}$ choose a rectangle of fixed width and height strictly inside the strip. On that rectangle the harmonic function $M_x - u$, with $M_x = C(1 + |x|)$ and C sufficiently large, is positive.

Its value at $(x, 0)$ is at most $C(1 + |x|)$ because $F(x) \geq 0$. Harnack's inequality, followed by interior derivative estimates, gives

$$|u(x, y)| + |\partial_x^j \partial_y^k u(x, y)| \leq C_{a', j, k}(1 + |x|), \quad |y| \leq a',$$

where for derivatives one first uses a slightly larger closed substrip. Thus $u(\cdot, y)$ and all derivatives needed below are tempered distributions, smoothly depending on y .

Use the Fourier convention $\hat{h}(\xi) = \int e^{-ix\xi} h(x) dx$. The harmonic equation becomes $\partial_y^2 \hat{u} = \xi^2 \hat{u}$. This equation is solved *locally in frequency*: multiply by a smooth function of compact support in ξ and solve the resulting distribution-valued ordinary differential equation. Multiplication by its fundamental matrix is then legitimate, since ξ remains in a compact set. The initial data are $\hat{u}(\xi, 0) = \hat{F}(\xi)$ and $\partial_y \hat{u}(\xi, 0) = 0$. Consequently

$$\hat{u}(\xi, a) = \cosh(a\xi) \hat{F}(\xi)$$

on every compact frequency interval. We do not multiply an arbitrary tempered distribution globally by the exponentially growing function $\cosh(a\xi)$.

The smooth function

$$m_a(\xi) = \frac{\xi^2}{\cosh(a\xi) - 1}, \quad m_a(0) = \frac{2}{a^2},$$

is Schwartz. The preceding local identity gives $m_a \hat{g}_a = -\xi^2 \hat{F}$ on every compact frequency interval, and hence as tempered distributions. To identify its inverse Fourier transform, the substitution $r = e^{2\pi x/a}$ and the beta integral give

$$\hat{h}_a(\xi) = \int_0^\infty \frac{r^{-ia\xi/(2\pi)}}{(1+r)^2} dr = \frac{a\xi/2}{\sinh(a\xi/2)}.$$

The value at zero is understood by continuity. It follows that $m_a = (2/a^2) \hat{h}_a^2$, proving (5.1). The convolution is an ordinary convergent integral because g_a has polynomial growth. Its distributional identity therefore holds pointwise.

Both h_a and J_a are strictly positive Schwartz functions. Moreover, $|h'_a| \leq c_a h_a$, so differentiation under the convolution gives $|J'_a| \leq c_a J_a$. Hence $J_a(x-t) \leq e^{c_a|x|} J_a(-t)$. Since $g_a \geq 0$, integration proves the first inequality in (5.2). Integrating twice, using $F(0) = F'(0) = 0$ and evenness, proves the second. $\square$

5.2 A zero-field exponential bound

Proposition 5.2. *Fix S and a sufficiently large finite β . There are $R_0 = R_0(S, \beta)$, $C_\beta < \infty$, and $\kappa > 0$ with the following property. Let $L \geq R \geq R_0$, with L even, and partition Λ_L into rectangular cells whose side lengths lie in $[R/2, 4R]$. If $f : \Lambda_L \rightarrow [-1, 1]$ has sum zero on every cell, then, for $q = R^{-5/2}$, $b = -\log(1 - q)$, and any spin component a ,*

$$\log \left\langle \exp \left(bx \sum_{z \in \Lambda_L} f_z S_z^a \right) \right\rangle \leq C_\beta bV, \quad x \in \mathbb{R}, \quad |x| \leq \kappa \log R. \quad (5.3)$$

Consequently there are $A_\beta, c > 0$ such that the spectral measurement of $D_f = \sum_z f_z S_z^a$ in the zero-field state satisfies

$$\mathbb{P} \left\{ |D_f| > \frac{A_\beta V}{\log R} \right\} \leq 2e^{-cbV}. \quad (5.4)$$

All bounds are uniform over the indicated partitions and tests.

Proof. It suffices first to take $a = 3$ and put the field in direction Y . Write $D_f = \sum_z f_z Z_z$ and define

$$F(z) = \frac{1}{bV} \log \langle e^{bzD_f} \rangle_b.$$

Since $[H_0, M_L^2] = 0$, cyclicity of the trace gives

$$\langle e^{bzD_f} \rangle_b = \frac{\text{Tr}_{\text{phys}}(e^{H_0} e^{bM_L^2/2} e^{bzD_f} e^{bM_L^2/2})}{\text{Tr}_{\text{phys}} e^{H_0 + bM_L^2}}.$$

The middle three factors form the restriction of a tensor product of single-slot sandwich matrices. Lemma 2.4, applied to $e^{bs^2/2} e^{bz f_z s^3} e^{bs^2/2}$, and Lemma 2.3 show that the expression inside the logarithm has no zeros in a fixed strip $|\text{Im } z| < a_0$. Here R_0 is large enough that $b \leq b_*$ from Lemma 2.4. Choose the analytic logarithm that is real on the real axis and vanishes at zero.

This is the logarithm of an *ordinary* Laplace transform: the spectral measure of D_f in the tilted Gibbs state is a probability measure, even though D_f need not commute with its density matrix. Rotation by π around the field axis makes this measure even. Therefore F is even, $F(x) \geq 0$, and, since $\|D_f\| \leq SV$,

$$\text{Re } F(x + iy) \leq F(x) \leq S|x|.$$

In particular,

$$F''(0) = \frac{b}{V} \text{Var}_b(D_f) = \frac{b}{V} \langle D_f^2 \rangle_b \leq C_\beta R^{-\eta}$$

by Proposition 4.1, with $\eta = 1/40$, and $b/q \leq 2$ for large R . This derivative is an ordinary variance, not the Duhamel covariance obtained by perturbing the Hamiltonian inside its exponential.

Fix $a \in (0, a_0)$ in Lemma 5.1. Choose $\kappa > 0$ so that $c_a \kappa < \eta/2$. Equation (5.2) then gives, uniformly over our tests,

$$0 \leq F(x) \leq C_\beta R^{-\eta/2}, \quad |x| \leq \kappa \log R.$$

To remove the field, let γ_0 and γ_b be the normalized zero-field and tilted density matrices. Since H_0 commutes with M_L^2 and $\|M_L^2\| \leq SV$,

$$\gamma_0 \leq e^{2bSV} \gamma_b.$$

Taking the trace against the positive operator e^{bxD_f} costs at most $2bSV$ in its logarithm and proves (5.3) for $a = 3$. Rotation invariance of γ_0 gives every common axis. Finally, exponential Markov at $x = \pm \kappa \log R$, with $\kappa A_\beta > C_\beta + c$, proves (5.4). $\square$

5.3 Comparison through nested partitions

For a rectangular cell C and an axis a , write

$$\bar{S}_C^a = \frac{1}{|C|} \sum_{x \in C} S_x^a.$$

All averages in a single fixed axis commute. The next proposition compares their ordinary squares; it makes no joint-measurement assertion about the three spin components.

Proposition 5.3. Fix S and a sufficiently large finite β . For every sufficiently large fixed R_* and all sufficiently large even L , there is a deterministic rectangular partition $\mathcal{P}_*(L)$ of Λ_L whose side lengths lie in $[R_*/2, 4R_*^{10/9}]$ and such that, for $a = 1, 2, 3$,

$$\frac{1}{V} \sum_{C \in \mathcal{P}_*(L)} |C| \langle (\bar{S}_C^a)^2 \rangle \leq \langle (M_L^a/V)^2 \rangle + \frac{C_\beta}{\log R_*} + o_L(1). \quad (5.5)$$

Here $o_L(1) \rightarrow 0$ at fixed R_* and β .

Proof. Set $R_0 = L$ and $R_{j+1} = R_j^{9/10}$, and stop at the largest n for which $R_n \geq R_*$. Starting with the one-cell partition, construct nested rectangular partitions $\mathcal{P}_j$ with cell sides in $[R_j/2, 4R_j]$. For example, subdivide each coordinate interval of a parent into nearly equal integer intervals of length comparable to R_{j+1} . The ratios R_j/R_{j+1} are large once R_* is large, so every such subdivision meets the stated bounds. The final scale satisfies $R_* \leq R_n < R_*^{10/9}$.

Fix an axis and perform its simultaneous site-spin measurement. Let $D(C)$ denote the measured value of $\bar{S}_C^a$. For one refinement, put $R = R_j$, $r = R_{j+1}$, and define

$$T_j = \frac{1}{V} \sum_{B \in \mathcal{P}_j} \sum_{\substack{C \in \mathcal{P}_{j+1} \\ C \subset B}} |C| |D(C) - D(B)|.$$

To express T_j through the tests of Proposition 5.2, assign a sign $\sigma_C \in \{-1, 1\}$ to every child and put

$$\bar{\sigma}_B = \frac{1}{|B|} \sum_{C \subset B} |C| \sigma_C, \quad f_x = \frac{\sigma_C - \bar{\sigma}_B}{2} \quad (x \in C \subset B).$$

Then $|f_x| \leq 1$ and $\sum_{x \in B} f_x = 0$. Direct expansion gives

$$\sum_x f_x S_x^a = \frac{1}{2} \sum_{B, C \subset B} |C| \sigma_C (D(C) - D(B))$$

in this measurement. Maximizing over the signs therefore gives $VT_j/2$. There are at most $\exp(CV/r^3)$ sign choices, because every child has volume at least $(r/2)^3$. Applying (5.4) to each test and taking a union bound yields

$$\mathbb{P}\{T_j > 2A_\beta/\log R\} \leq 2 \exp\{CVR^{-27/10} - cbV\} \leq 2 \exp(-c'VR^{-5/2}).$$

The last inequality holds uniformly after increasing R_* , since $b \geq R^{-5/2}$ and $R^{-27/10} = o(R^{-5/2})$.

As $|D(C) - D(B)| \leq 2S$, this also bounds the expected weighted square increment:

$$\mathbb{E} \left[\frac{1}{V} \sum_{B, C \subset B} |C| (D(C) - D(B))^2 \right] \leq \frac{C_\beta}{\log R_j} + C \exp(-c'VR_j^{-5/2}).$$

The squares telescope exactly, because $D(B)$ is the weighted mean of its children's averages:

$$\sum_{C \in \mathcal{P}_{j+1}} |C| D(C)^2 - \sum_{B \in \mathcal{P}_j} |B| D(B)^2 = \sum_{B, C \subset B} |C| (D(C) - D(B))^2.$$

Now $\sum_{j < n} (\log R_j)^{-1} \leq C/\log R_*$, whereas $VR_j^{-5/2} \geq L^{1/2}$ and $n = O(\log \log L)$. Summing the expected increments proves (5.5) with $\mathcal{P}_*(L) = \mathcal{P}_n$. The deterministic construction works for all three axes. $\square$

6 Equilibrium states and vanishing replica cross terms

The spatial estimate has a second use besides comparing block averages: it removes certain couplings between independent replicas at a pressure tangent. We give the variational framework first, including the precise order of limits needed for this deduction. The underlying lattice is now $\mathbb{Z}^3$, with the spin- S matrix algebra at each site. A state is a consistent family of density matrices on finite sets, or equivalently a positive normalized functional on the norm closure of the local matrix algebras. Local convergence means convergence on every local observable. Write $\mathcal{S}_{\text{TI}}$ for the compact convex set of translation-invariant states, and τ_x for translation by x .

6.1 The variational principle and its tangents

For a density matrix ρ , let $S(\rho) = -\text{Tr}(\rho \log \rho)$. If $Q_n = \{0, \dots, n-1\}^3$ and $\omega \in \mathcal{S}_{\text{TI}}$, its mean entropy and unperturbed interaction expectation per site are

$$\mathfrak{s}(\omega) = \lim_{n \rightarrow \infty} \frac{S(\omega_{Q_n})}{|Q_n|}, \quad e_0(\omega) = \beta \sum_{r=1}^3 \omega(\mathbf{S}_0 \cdot \mathbf{S}_{e_r}).$$

For a self-adjoint local observable a , let $A_L = \sum_{x \in \Lambda_L} \tau_x(a)$, interpreted periodically for L larger than its support, and set

$$P_a(t) = \lim_{L \rightarrow \infty} \frac{1}{V} \log \text{Tr} e^{H_0 + tA_L}, \quad t \in \mathbb{R}.$$

The limit uses even L , as elsewhere. Finite range and boundedness make free and periodic boundary conditions equivalent for this limit: the difference of their exponents has norm $O(L^2)$.

The following standard variational facts are recalled with proofs to specify the state space on which the later tangent is taken; see [10, Theorems 2–3] and [11, Theorem 3].

Proposition 6.1. *The mean entropy exists on $\mathcal{S}_{\text{TI}}$, is upper semicontinuous and affine, and*

$$P_a(t) = \max_{\omega \in \mathcal{S}_{\text{TI}}} \{\mathfrak{s}(\omega) + e_0(\omega) + t\omega(a)\}. \quad (6.1)$$

In particular, writing $P_0 = \beta p_{S,\beta}(0)$, the set

$$\mathcal{E}_\beta = \{\omega \in \mathcal{S}_{\text{TI}} : \mathfrak{s}(\omega) + e_0(\omega) = P_0\}$$

is a nonempty compact face of $\mathcal{S}_{\text{TI}}$. Every local limit of periodic zero-field Gibbs states belongs to $\mathcal{E}_\beta$, and

$$P'_a(0+) = \max_{\omega \in \mathcal{E}_\beta} \omega(a). \quad (6.2)$$

The extreme points of $\mathcal{E}_\beta$ are precisely its states that are extreme in $\mathcal{S}_{\text{TI}}$; we call these states ergodic.

Proof. Tile a large cube by translates of Q_n and a remainder of vanishing relative volume. Entropy subadditivity and the bound $S(\rho_B) \leq |B| \log(2S+1)$ show that

$$\mathfrak{s}(\omega) = \inf_{n \geq 1} \frac{S(\omega_{Q_n})}{|Q_n|}.$$

This proves existence and upper semicontinuity, since each finite-volume entropy is continuous. For a mixture $\rho = \lambda \rho_1 + (1-\lambda) \rho_2$, the entropy lies between $\lambda S(\rho_1) + (1-\lambda) S(\rho_2)$ and this quantity plus the binary entropy of λ . Division by volume proves affinity of $\mathfrak{s}$.

Apply the finite-dimensional Gibbs variational principle to the restriction of any $\omega \in \mathcal{S}_{\text{TI}}$ to a large cube. Boundary terms are $o(V)$ in norm, so $P_a(t) \geq \mathfrak{s}(\omega) + e_0(\omega) + t\omega(a)$. Conversely, choose a locally convergent subsequence of periodic Gibbs states for $H_0 + tA_L$, with limit ω . The limit is translation invariant, and local interaction expectations converge. Tiling the large cubes by fixed cubes Q_n gives

$$\limsup_{L \rightarrow \infty} \frac{S(\gamma_{L,t})}{V} \leq \frac{S(\omega_{Q_n})}{|Q_n|} \quad \text{for every } n.$$

Letting $n \rightarrow \infty$ and using the exact finite-volume Gibbs identity gives the reverse variational inequality. At $t = 0$ this argument also proves the assertion about every periodic Gibbs local limit. Compactness, upper semicontinuity, and affinity show that $\mathcal{E}_\beta$ is a nonempty compact face; the assertion about its extreme points follows from the definition of a face.

For completeness, a maximizer ω_t in (6.1), with $t > 0$, satisfies

$$\max_{\omega \in \mathcal{E}_\beta} \omega(a) \leq \frac{P_a(t) - P_0}{t} \leq \omega_t(a).$$

As $t \downarrow 0$, upper semicontinuity shows that every local limit of ω_t belongs to $\mathcal{E}_\beta$: its unperturbed variational value tends to P_0 , since $|\omega_t(a)| \leq \|a\|$. Taking a convergent subsequence proves (6.2). $\square$

In the field direction Y , the exponent-normalized pressure is $\beta p_{S,\beta}(b/\beta)$. Thus

$$m = \max_{\omega \in \mathcal{E}_\beta} \omega(Y_0). \quad (6.3)$$

Rotation invariance gives the same maximum in every unit direction. Consequently the vector $\mathbf{m}(\omega) = (\omega(X_0), \omega(Y_0), \omega(Z_0))$ has length at most m for every $\omega \in \mathcal{E}_\beta$. The face maximizing $\omega(Y_0)$ has an extreme point, so an ergodic equilibrium state attains m . Section 7 will prove that every ergodic equilibrium state has this same magnetization magnitude.

The following mean-ergodic consequence of extremality is standard for asymptotically commuting lattice translations; see [9, Theorem 3].

Lemma 6.2 (Ergodic averages). *If ω is an extreme translation-invariant state and A is a self-adjoint local observable, then*

$$\lim_{n \rightarrow \infty} \omega \left[\left(\frac{1}{|Q_n|} \sum_{x \in Q_n} \tau_x(A) - \omega(A)\mathbf{1} \right)^2 \right] = 0.$$

Proof. After shifting and scaling, assume $0 \leq A \leq \mathbf{1}$, and denote its average by $\bar{A}_n$. In the Hilbert space of the state, translations are commuting unitaries. The mean ergodic theorem makes $\bar{A}_n \Omega$ converge to a vector η , where Ω represents the state. Hence $\lambda(C) = \lim_n \omega(C \bar{A}_n)$ exists for every local C .

For local $C \geq 0$, locality gives $\|[C^{1/2}, \bar{A}_n]\| \rightarrow 0$. Therefore $\lambda(C) = \lim_n \omega(C^{1/2} \bar{A}_n C^{1/2})$, and $0 \leq \lambda(C) \leq \omega(C)$. The functional extends continuously, is translation invariant by the boundary-to-volume estimate for the averages, and satisfies $\lambda(\mathbf{1}) = \omega(A)$. Extremality of ω now gives $\lambda = \omega(A)\omega$: otherwise its normalized parts in $\omega = \lambda + (\omega - \lambda)$ would give a nontrivial convex decomposition into translation-invariant states. Testing on the dense set of local vectors identifies $\eta = \omega(A)\Omega$, which is the asserted variance convergence. $\square$

6.2 Products of zero-sum observables in distinct replicas

Take j copies of the spin system, with superscripts $[r]$ denoting the replica. The unperturbed exponent is $\sum_{r=1}^j H_0^{[r]}$, and its pressure is jP_0 . Every translation-invariant equilibrium state for this interaction has each marginal in $\mathcal{E}_\beta$. Indeed, entropy subadditivity between replicas bounds its variational value by the sum of the marginal variational values, each at most P_0 ; equality of the total forces equality for each marginal. Conversely, products of states in $\mathcal{E}_\beta$ are replicated equilibrium states, because their entropies and interaction expectations add.

For real coefficients v_i on finitely many qubit slots, define

$$D_v^a = \sum_i v_i s_i^a, \quad B_v^a = P_* D_v^a P_*$$

as a physical local operator. On a site's symmetric subspace, $P_* s_{(x,k)}^a P_* = S_x^a / \ell$. Thus

$$B_v^a = \sum_x c_x S_x^a, \quad c_x = \frac{1}{\ell} \sum_{k=1}^\ell v_{(x,k)}, \quad \sum_x c_x = 0 \quad \text{if} \quad \sum_i v_i = 0. \quad (6.4)$$

The next lemma concerns products of these *compressed* operators in different replicas. It does not identify $P_*(D_v^a)^2 P_*$ with $(B_v^a)^2$.

Lemma 6.3 (Replica cross terms). *Fix S and a sufficiently large finite β . Let ω be any translation-invariant equilibrium state of j unperturbed replicas. For distinct replicas r, s , axes a, b , and finitely supported real slot coefficients v, w with $\sum_i v_i = \sum_i w_i = 0$,*

$$\omega((B_v^a)^{[r]} (B_w^b)^{[s]}) = 0.$$

The same holds for every finite linear combination of translates of such products.

Proof. Write $O = (B_v^a)^{[r]} (B_w^b)^{[s]}$ and choose a fixed cube of side d containing both spatial supports. Let ρ_L be the restriction of ω to a cube of side L , and let γ_L be the product of the j periodic zero-field Gibbs density matrices on that cube. With $D(\rho \parallel \gamma) = \text{Tr } \rho(\log \rho - \log \gamma)$,

$$\frac{1}{V} D(\rho_L \parallel \gamma_L) \longrightarrow 0. \quad (6.5)$$

Indeed, expanding $\log \gamma_L$ gives j times the finite-volume pressure minus the entropy density and interaction density of ρ_L . Their limits cancel by the equilibrium variational identity; periodic bonds at the boundary change the interaction by $o(V)$.

Fix an integer $R \geq d$ and take L through even multiples of R . Partition the cube into cells of side R , and let $\mathcal{T}$ be the translations of the chosen support cube lying wholly within individual cells. In particular,

$$\frac{|\mathcal{T}|}{V} = \left(\frac{R - d + 1}{R} \right)^3.$$

Write $B_v^a = \sum_x c_x S_x^a$ and $B_w^b = \sum_x d_x S_x^b$, as in (6.4). Choose $C_O \geq \max\{1, S \|c\|_1 \|d\|_1\}$ and put

$$O' = \frac{1}{C_O} \sum_{y \in \mathcal{T}} \tau_y(O).$$

All operators in this sum can be measured simultaneously, using axis a in replica r and axis b in replica s . Conditional on any outcome in replica s , denote the value of $\tau_y(B_w^b)^{[s]}$ by b_y , so $|b_y| \leq S\|d\|_1$. As a function of the first replica's outcomes, O' is then $\sum_z f_z S_z^a$, where

$$f_z = \frac{1}{C_O} \sum_{y \in \mathcal{T}} c_{z-y} b_y.$$

These coefficients have absolute value at most one. Their sum in each cell vanishes, since each translated support stays in that cell and $\sum_x c_x = 0$. Independence of the replicas under γ_L and Proposition 5.2, applied conditionally, therefore give

$$\log \text{Tr}(\gamma_L e^{\pm \theta O'}) \leq C_\beta bV, \quad b = -\log(1 - R^{-5/2}), \quad \theta = \kappa b \log R. \quad (6.6)$$

For any self-adjoint K , the Gibbs variational principle followed by the Golden–Thompson inequality [8, 20] gives

$$\text{Tr}(\rho_L K) \leq D(\rho_L \|\gamma_L) + \log \text{Tr} e^{\log \gamma_L + K} \leq D(\rho_L \|\gamma_L) + \log \text{Tr}(\gamma_L e^K).$$

Use $K = \theta O'$ or $K = -\theta O'$ according to the sign of $\omega(O)$. Translation invariance and (6.6) yield

$$\theta \frac{|\mathcal{T}|}{C_O} |\omega(O)| \leq D(\rho_L \|\gamma_L) + C_\beta bV.$$

First let $L \rightarrow \infty$ at this fixed R ; this order is required because the entropy density in (6.5) has no rate. For large R , $|\mathcal{T}|/V$ is bounded below, and we obtain $|\omega(O)| \leq C_{\beta,O}/\log R$. Letting $R \rightarrow \infty$ proves the claim. Linearity gives the final assertion. $\square$

Thus, for a finite-range replica perturbation that is a sum of single-replica terms and the cross terms in Lemma 6.3, its pressure tangent depends only on the single-replica terms. The construction in the next section uses this fact to study negative squares of block magnetizations.

7 Pressure tangents and the equilibrium magnetization radius

The variational principle bounds the magnetization of every state in $\mathcal{E}_\beta$ by m . We now prove that every ergodic member has magnetization of length exactly m . The obstruction to a smaller radius will be analyticity of certain pressure tangents. To obtain that analyticity, we construct random transfer matrices whose traces are nonzero before taking expectations; Lemma 6.3 then identifies the pressure tangents of their moments.

For an integer $r \geq 2$, write $C_r = \{0, \dots, r-1\}^3$, and let $\mathcal{C}_r$ denote all its translates, on the lattice or on a sufficiently large torus. For any such cube C , put $X_C = \sum_{x \in C} X_x$, and define Y_C, Z_C in the same way. Fix $B \geq 2$ and $W \geq 2B$. We will choose a constant $D_B > 0$, independent of W , and consider the nonpositive local observable

$$a_{B,W} = -\frac{Y_{C_B}^2}{B^6} - D_B \frac{X_{C_W}^2 + Z_{C_W}^2}{W^6}. \quad (7.1)$$

In an ergodic state with magnetization vector $(0, r, 0)$, the transverse term tends to zero as $W \rightarrow \infty$ at fixed B , while the Y term tends to $-r^2$ as $B \rightarrow \infty$, by Lemma 6.2. The periodic sum of $a_{B,W}$ is denoted by $A_{B,W,L} = \sum_{x \in \Lambda_L} \tau_x(a_{B,W})$. Define

$$s_{B,W}(x) = \max_{\omega \in \mathcal{E}_\beta} \{\omega(a_{B,W}) + x\omega(Y_0)\}, \quad x \in \mathbb{R}. \quad (7.2)$$

This function is convex and S -Lipschitz. It is even: a rotation by π around the X -axis preserves $a_{B,W}$ and sends Y_0 to $-Y_0$.

Proposition 7.1 (Harmonic pressure tangent). *For every integer $B \geq 2$ there is $D_B > 0$ such that, for every integer $W \geq 2B$, the function in (7.2) has a harmonic extension $v_{B,W}$ from the positive real axis to $\mathbb{H} = \{z \in \mathbb{C} : \Re z > 0\}$ satisfying*

$$v_{B,W}(x) = s_{B,W}(x) \quad (x > 0), \quad v_{B,W}(z) \leq s_{B,W}(\Re z) \quad (z \in \mathbb{H}). \quad (7.3)$$

We prove the proposition in the next three subsections. We first express the negative squares in (7.1) as infinitesimal contributions of real single-slot matrices, together with a small reduction of the ferromagnetic coupling. Shared Gaussian coefficients then turn moments of the random traces into replicated pressures. Finally a second-moment argument selects nonvanishing traces whose logarithms converge to the required harmonic function.

7.1 A Laplacian construction for the negative squares

Recall that there are $\ell = 2S$ slots per site and that P_* projects onto the physical spin space. For a real vector $v = (v_i)$ on finitely many slots, write

$$D_v^a = \sum_i v_i s_i^a.$$

Our generators will be iD_v^2 and D_v^1, D_v^3 , with $\sum_i v_i = 0$. All three are real matrices in the s^3 basis, and their exponentials with real coefficients are products of single-slot matrices in $SL_2(\mathbb{R})$. Squaring iD_v^2 contributes a negative square in the Y -direction. The other two generators contribute positive squares; the reduction of the static coupling will cancel their unwanted nearest-neighbor coefficients.

All quadratic coefficients below use *ordered* slot pairs:

$$(D_v^a)^2 = \sum_{i,j} v_i v_j s_i^a s_j^a.$$

For a cube C , let L_C^{full} be the graph Laplacian of the complete graph on its $\ell|C|$ slots. Let L_C^{bond} be the Laplacian with an edge between every pair of slots whose sites are nearest neighbors inside C . Thus an off-diagonal Laplacian entry is -1 on an edge. These matrices are extended by zero when used on a larger slot set.

The bond graph of C_B is connected. Consequently one can choose $c_B > 0$ so that, for every $C \in \mathcal{C}_B$,

$$Q_C = c_B L_C^{\text{bond}} - B^{-6} L_C^{\text{full}} \geq 0. \quad (7.4)$$

Indeed, both matrices annihilate the constant vector, and on its orthogonal complement the bond Laplacian is positive definite while $L_C^{\text{full}} = \ell B^3 I$. Decompose Q_C spectrally as $\sum_v v v^\top$. Every vector in this sum is real and has zero sum; use the generators iD_v^2 for this list.

To cancel the remaining static reduction in the X, Z directions, put $n_{B,W} = (W - B + 1)^3$ and, for $C \in \mathcal{C}_W$, consider

$$R_C = D_B W^{-6} L_C^{\text{full}} - \frac{c_B}{n_{B,W}} \sum_{\substack{C' \in \mathcal{C}_B \\ C' \subseteq C}} L_{C'}^{\text{bond}}. \quad (7.5)$$

The constant D_B can be chosen so that $R_C \geq 0$ for every $W \geq 2B$. Here is a uniform bound. Each slot belongs to at most B^3 small cubes, and its degree in each bond graph is at most 6ℓ . The sum of the bond Laplacians therefore has norm at most $12\ell B^3$. Since $n_{B,W} \geq W^3/8$, the

second term in (7.5) has norm at most $96c_B\ell B^3/W^3$. On the zero-sum subspace, the first term is $D_B\ell/W^3$ times the identity. Thus, for example,

$$D_B = 96c_B B^3 + 1 \quad (7.6)$$

suffices. Decompose each R_C as $\sum_v vv^\top$ with real zero-sum vectors, and use both generators D_v^1 and D_v^3 . Choose the lists on one cube of each size and translate them. The resulting families are translation invariant.

Let $T_{B,L}$ be the physical compression of the static reduction

$$-c_B \sum_{C \in \mathcal{C}_B} \sum_{\substack{i,j \text{ slots in } C \\ x_i, x_j \text{ nearest neighbors}}} \sum_{a=1}^3 s_i^a s_j^a, \quad (7.7)$$

where the inner sum is ordered. Each unordered site bond belongs to $(B-1)B^2$ translates of C_B , so

$$H_{0,t} := H_0 + tT_{B,L} = (\beta - 2tc_B(B-1)B^2) \sum_{\{x,y\} \text{ bond}} \mathbf{s}_x \cdot \mathbf{s}_y.$$

For fixed B this remains ferromagnetic for all sufficiently small $t > 0$, uniformly in $L > 2W$.

Denote all the generators just constructed by G_α . Their quadratic contribution is compressed *after squaring*:

$$T_{B,L} + \sum_{\alpha} P_* G_\alpha^2 P_* = A_{B,W,L} + C_{B,W} VI \quad (7.8)$$

on physical space, for a real scalar $C_{B,W}$ independent of $L > 2W$. To check the identity, first consider distinct slots i, j in the cube C under consideration. If $E_C(i, j)$ is the indicator of an edge of the bond graph, the coefficients from a small cube in the Y -direction are

$$\underbrace{-c_B E_C(i, j)}_{\text{static reduction}} + \underbrace{(c_B E_C(i, j) - B^{-6})}_{-(Q_C)_{ij}} = -B^{-6}.$$

For either the X - or the Z -direction, the contribution from a large cube is

$$(R_C)_{ij} = -D_B W^{-6} + \frac{c_B}{n_{B,W}} \sum_{\substack{C' \in \mathcal{C}_B \\ C' \subseteq C}} E_{C'}(i, j).$$

Every small cube lies in exactly $n_{B,W}$ large cubes. After summing over translates, the second term cancels the static reduction, leaving the coefficient $-D_B W^{-6}$ on each ordered pair in each large cube. These are precisely the off-diagonal coefficients of $A_{B,W,L}$. The remaining terms have $i = j$ and are scalar because $(s_i^a)^2 = I/4$; translation invariance gives the scalar in (7.8). In particular, the identity does not replace $P_* G_\alpha^2 P_*$ by $(P_* G_\alpha P_*)^2$, which would in general give different single-replica terms.

7.2 Nonvanishing random traces and replicated moments

Fix B, W and a sufficiently small $t > 0$. For an even $L > 2W$ and an integer $k \geq 1$, take independent real centered Gaussian variables $\lambda_{r,\alpha}$, $1 \leq r \leq k$, of variance $2t/k$. Fix an order of the generators and define a slot-space layer

$$U_r(z) = \left(\prod_{\alpha} e^{\lambda_{r,\alpha} G_{\alpha}} \right) e^{(tz/k) \sum_i s_i^2}.$$

Only the complete layer is compressed. With all products taken in their fixed order, define the normalized random entire function

$$\mathcal{Z}_{L,k,t}(z) = \frac{e^{-tC_{B,W}V}}{\text{Tr } e^{H_0}} \text{Tr}_{\text{phys}} \prod_{r=1}^k \left[e^{H_{0,t}/k} (P_* U_r(z) P_*) \right]. \quad (7.9)$$

Gaussian exponential moments ensure that all finite products of these functions are integrable, locally uniformly in their arguments.

Every realization of (7.9) is nonzero on $\mathbb{H}$. Although the pulse supports overlap, their fixed ordered product is still a tensor product of single-slot matrices: factors on different slots commute and can be regrouped without changing the order on any one slot. All pulse matrices on a slot lie in $SL_2(\mathbb{R})$ and hence act as disk automorphisms in the s^2 basis, by Lemma 2.4. On every slot the remaining factor $e^{tzs^2/k}$ sends the closed disk strictly inside the disk when $\Re z > 0$. Their composition therefore has the strict mapping property of Lemma 2.3. To identify the permutation layers, write E for the full-slot exponential of the static interaction with time step $1/k$. It commutes with P_* , and cyclicity gives

$$\text{Tr}_{\text{phys}} \prod_{r=1}^k [e^{H_{0,t}/k} (P_* U_r(z) P_*)] = \text{Tr}_{\text{slots}} \prod_{r=1}^k [(EP_*) U_r(z)].$$

Each EP_* has nonnegative exchange rates and a symmetric seam, so it is a permutation layer of the class allowed in that lemma. The identity uses $P_*^2 = P_*$ and does not require U_r to preserve physical space. This proves samplewise nonvanishing, and in particular

$$z \mapsto \frac{1}{tV} \log |\mathcal{Z}_{L,k,t}(z)| \quad \text{is harmonic on } \mathbb{H}. \quad (7.10)$$

Complex conjugation has a specific sign here. The static matrices, the projections and all pulses are real in the s^3 basis, whereas s^2 is purely imaginary. Therefore, for the *same* Gaussian sample,

$$\overline{\mathcal{Z}_{L,k,t}(z)} = \mathcal{Z}_{L,k,t}(-\bar{z}). \quad (7.11)$$

The argument on the right may lie outside $\mathbb{H}$; the function is entire, so this causes no difficulty. In particular its second moment uses the pair $(z, -\bar{z})$, not $(z, \bar{z})$.

We next compute products of traces using replicas, always sharing the Gaussian variables between replicas. For fixed j and arguments $z_1, \dots, z_j \in \mathbb{C}$, the finite-dimensional product formula gives

$$\begin{aligned} \lim_{k \rightarrow \infty} \mathbb{E} \prod_{r=1}^j \mathcal{Z}_{L,k,t}(z_r) &= \mathcal{R}_{L,t}^{(j)}(z_1, \dots, z_j) \\ &:= \frac{\text{Tr} \exp \left\{ \sum_{r=1}^j (H_0^{[r]} + tA_{B,W,L}^{[r]} + tz_r M_L^{2,[r]} + tA_{\text{cross},L}^{(j)}) \right\}}{(\text{Tr } e^{H_0})^j}. \end{aligned} \quad (7.12)$$

The limit is locally uniform in the arguments. To verify both the limit and its cross terms, let $P^{(j)}$ be the product of the physical projections in all replicas. For one shared Gaussian variable,

$$\mathbb{E} \exp \left(\lambda \sum_{r=1}^j G_{\alpha}^{[r]} \right) = \exp \left(\frac{t}{k} \left(\sum_{r=1}^j G_{\alpha}^{[r]} \right)^2 \right).$$

Independence allows these expectations to be taken in the prescribed pulse order. After the outer compression, the first-order term is the following operator on physical replica space:

$$P^{(j)} \left(\sum_{r=1}^j G_\alpha^{[r]} \right)^2 P^{(j)} = \sum_{r=1}^j (P_* G_\alpha^2 P_*)^{[r]} + 2 \sum_{r < s} (P_* G_\alpha P_*)^{[r]} (P_* G_\alpha P_*)^{[s]}. \quad (7.13)$$

For an imaginary-direction generator $G_\alpha = iD_v^2$, the cross product on the right is

$$-(P_* D_v^2 P_*)^{[r]} (P_* D_v^2 P_*)^{[s]}.$$

For the other two directions it has the positive sign. Thus

$$A_{\text{cross}, L}^{(j)} = 2 \sum_{\alpha} \sum_{r < s} (P_* G_\alpha P_*)^{[r]} (P_* G_\alpha P_*)^{[s]}$$

is self-adjoint, finite range and translation invariant. Every one of its local summands is a real multiple of a cross term in Lemma 6.3, since all the vectors v have zero sum.

Using (7.8) and distributing the scalar correction in (7.9) among the k steps, the expected replicated step is $I + k^{-1} \mathcal{H} + O(k^{-2})$, where $\mathcal{H}$ is the exponent in (7.12). Here B, W, j, t, L are fixed and the error is locally uniform in $(z_1, \dots, z_j)$. Independence between time steps makes the expected product the k th power of this deterministic step. The matrix product limit [22] proves (7.12).

For real $x_1, \dots, x_j$, the numerator defining $\mathcal{R}_{L,t}^{(j)}$ is positive. The thermodynamic limit followed by the right pressure tangent gives

$$\lim_{t \downarrow 0} \lim_{L \rightarrow \infty} \frac{1}{tV} \log \mathcal{R}_{L,t}^{(j)}(x_1, \dots, x_j) = \sum_{r=1}^j s_{B,W}(x_r). \quad (7.14)$$

Indeed, by (6.2), the left side is the maximum of the perturbation density over equilibrium states of the unperturbed replicated interaction. Lemma 6.3 makes the cross density vanish in every such state. Each marginal belongs to $\mathcal{E}_\beta$, giving the upper bound by the sum on the right. The product of maximizing single-replica states is an equilibrium state and attains that sum. This proves equality.

We also need an upper bound at complex arguments. For self-adjoint matrices A, B ,

$$|\text{Tr } e^{A+iB}| \leq \text{Tr } e^A. \quad (7.15)$$

For completeness, apply the product formula to $(e^{A/n} e^{iB/n})^n$ and then Schatten Hölder: its trace norm is at most $\|e^{A/n}\|_n^n = \text{Tr } e^A$, since the intervening factors are unitary. Taking the limit proves the inequality. Applied to (7.12), it bounds its absolute value by $\mathcal{R}_{L,t}^{(j)}(\Re z_1, \dots, \Re z_j)$.

The limits in (7.14) are uniform on compact sets of real arguments in their stated order. Indeed the normalized log traces are S -Lipschitz in each real argument, uniformly in L, t , by the norm bound $\|M_L^2\| \leq SV$; finite nets promote pointwise convergence to uniform convergence. The same bound holds for their limits. Combining this observation with (7.11), (7.12) and (7.15), we obtain the following useful finite parameter statement. Given a compact $K \subset \mathbb{H}$, finitely many positive real points $x_1, \dots, x_q$, and $\epsilon > 0$, one can choose $t > 0$ sufficiently small, then an even L sufficiently

large, and finally k sufficiently large, so that, with $T = tV$ and $\mathcal{Z} = \mathcal{Z}_{L,k,t}$,

$$\mathbb{E}|\mathcal{Z}(z)|^2 \leq \exp\{T(2s_{B,W}(\Re z) + \epsilon)\}, \quad z \in K, \quad (7.16)$$

$$\left| T^{-1} \log \mathbb{E}U - 2 \sum_{i=1}^q s_{B,W}(x_i) \right| \leq \epsilon, \quad (7.17)$$

$$\left| T^{-1} \log \mathbb{E}U^2 - 4 \sum_{i=1}^q s_{B,W}(x_i) \right| \leq \epsilon, \quad U = \prod_{i=1}^q |\mathcal{Z}(x_i)|^2. \quad (7.18)$$

The parameter T can be required to exceed any prescribed number, by enlarging L after t has been fixed. For the last two estimates, use the real argument lists $(x_1, -x_1, \dots, x_q, -x_q)$ and twice that list, together with the evenness of $s_{B,W}$. These products are nonnegative random variables at finite k . Their limits at fixed t, L are strictly positive, so the locally uniform product limit also gives the logarithmic estimates. For the first estimate it gives the required upper bound after the finite-volume parameters have been fixed, even when its exponential right side is small.

This order of choices is essential: the matrix product limit is taken at fixed volume; the pressure limit then precedes differentiation at $t = 0$. The finite estimates are implemented in the reverse order, choosing t , then L , then k . No zero-free assertion about an averaged partition function has been used.

7.3 Selecting harmonic logarithms

We isolate the analytic step to explain how moment information retains the samplewise nonvanishing just proved.

Lemma 7.2 (Selection from moment bounds). *Let $s : (0, \infty) \rightarrow \mathbb{R}$ be locally Lipschitz, and enumerate a countable dense subset $D = \{x_1, x_2, \dots\}$ of $(0, \infty)$ with $x_1 = 1$. Suppose F_n are random holomorphic functions that are almost surely nonzero on $\mathbb{H}$, and $T_n \rightarrow \infty$. Assume, locally uniformly for $z \in \mathbb{H}$, that*

$$\limsup_{n \rightarrow \infty} \left\{ T_n^{-1} \log \mathbb{E}|F_n(z)|^2 - 2s(\Re z) \right\} \leq 0. \quad (7.19)$$

For every $q \geq 1$, suppose also that, with $U_n = \prod_{i=1}^q |F_n(x_i)|^2$,

$$T_n^{-1} \log \mathbb{E}U_n \rightarrow 2 \sum_i s(x_i), \quad T_n^{-1} \log \mathbb{E}U_n^2 \rightarrow 4 \sum_i s(x_i). \quad (7.20)$$

Then there is a harmonic function v on $\mathbb{H}$ with $v(x) = s(x)$ for every $x > 0$ and $v(z) \leq s(\Re z)$ everywhere.

Proof. Fix $q \geq 1$, a compact $K \subset \mathbb{H}$ containing $x_1, \dots, x_q$, and $\delta > 0$. Put $h_n = T_n^{-1} \log |F_n|$. The Paley–Zygmund inequality [16] and (7.20) give, for each $\epsilon > 0$ and all sufficiently large n ,

$$\mathbb{P}\{U_n \geq \tfrac{1}{2} \mathbb{E}U_n\} \geq \frac{(\mathbb{E}U_n)^2}{4\mathbb{E}U_n^2} \geq \tfrac{1}{4} e^{-3\epsilon T_n}. \quad (7.21)$$

We compare this with the probability that h_n exceeds $s(\Re z) + \delta$ somewhere on K .

Cover K by finitely many disks D_j with concentric doubled disks $\tilde{D}_j$ compactly contained in $\mathbb{H}$, chosen so that the oscillation of $s(\Re z)$ on each $\tilde{D}_j$ is at most $\delta/4$. Write s_j for its minimum there. The submean inequality for the nonnegative subharmonic function $|F_n|^2$ gives

$$\sup_{D_j} |F_n|^2 \leq C_j \int_{\tilde{D}_j} |F_n(z)|^2 dA(z).$$

Using (7.19), the expectation of the right side is at most $C'_j \exp\{T_n(2s_j + \delta/2 + \epsilon)\}$ for large n . If $h_n(z) > s(\Re z) + \delta$ at any point of D_j , then its left side exceeds $\exp\{T_n(2s_j + 2\delta)\}$. Markov's inequality and a sum over the disks therefore yield

$$\mathbb{P}\{h_n(z) > s(\Re z) + \delta \text{ for some } z \in K\} \leq C_K e^{-(3\delta/2 - \epsilon)T_n}. \quad (7.22)$$

Choose $\epsilon > 0$ with $4\epsilon < 3\delta/2$. Since $T_n \rightarrow \infty$, the bound in (7.22) is eventually smaller than the lower bound in (7.21). There is therefore a realization for which both

$$h_n(z) \leq s(\Re z) + \delta \quad (z \in K), \quad \sum_{i=1}^q h_n(x_i) \geq \sum_{i=1}^q s(x_i) - \frac{\epsilon}{2} - \frac{\log 2}{2T_n}$$

hold. Subtracting the upper bounds at the other $q - 1$ points gives

$$s(x_i) - (q - 1)\delta - \frac{\epsilon}{2} - \frac{\log 2}{2T_n} \leq h_n(x_i) \leq s(x_i) + \delta. \quad (7.23)$$

Exhaust $\mathbb{H}$ by compact rectangles, and at stage q enlarge the rectangle to include the first q points. Apply the preceding construction with $\delta = q^{-2}$ and with ϵ small enough as above, choosing an increasing sequence of indices. The selected harmonic functions are locally bounded above, and their values converge to s at each point of D . Their values at 1 in particular remain bounded below. On any connected relatively compact subdomain containing 1, subtracting these functions from a common upper bound and using Harnack's inequality gives local lower bounds as well. Interior estimates and a diagonal subsequence yield a locally uniform harmonic limit v . The compact upper bounds give $v(z) \leq s(\Re z)$; the values on D , followed by continuity, give $v(x) = s(x)$ for every $x > 0$. $\square$

Completion of the proof of Proposition 7.1. Fix B, W and enumerate a countable dense set of positive real points, starting with 1. Apply (7.16)–(7.18) successively on an exhaustion of $\mathbb{H}$, with errors tending to zero and with $T_n = t_n L_n^3 \geq n$. At stage n require the product estimates for the first q points for each $q \leq n$. This is a finite collection of requirements, so the same parameter choices can enforce all of them. The resulting random functions $F_n = \mathcal{Z}_{L_n, k_n, t_n}$ satisfy Lemma 7.2, by samplewise nonvanishing and the moment estimates. Its conclusion is precisely (7.3). $\square$

7.4 All ergodic equilibrium states have the same radius

The negative square in (7.1) now separates two hypothetical magnetization radii: a state with smaller radius would dominate at small positive x , while the largest radius must dominate at large x . Harmonicity prevents this change.

Proposition 7.3 (Equilibrium magnetization radius). *Every ergodic translation-invariant state $\omega \in \mathcal{E}_\beta$ satisfies*

$$\omega(X_0)^2 + \omega(Y_0)^2 + \omega(Z_0)^2 = m^2. \quad (7.24)$$

Proof. The maximum of $\omega(Y_0)$ over the compact face $\mathcal{E}_\beta$ is m . The maximizing states form a nonempty compact face, so it contains an extreme point ω_m of the translation-invariant state space. Thus ω_m is ergodic. The bound on the length of every equilibrium magnetization vector implies that its vector is $(0, m, 0)$.

Suppose an ergodic equilibrium state has magnetization length $m' < m$. Rotate it to a state $\omega_{m'}$ with vector $(0, m', 0)$; rotations preserve equilibrium and translation ergodicity. By Lemma 6.2, for $r = m, m'$,

$$\omega_r(Y_{C_B}^2/B^6) \longrightarrow r^2, \quad \omega_r((X_{C_W}^2 + Z_{C_W}^2)/W^6) \longrightarrow 0.$$

For each B , choose $W(B) \geq 2B$ so large that the second expectation, multiplied by D_B , is at most $1/B$ in both states. Then, writing $s_B = s_{B,W(B)}$,

$$\liminf_{B \rightarrow \infty} s_B(x) \geq \max\{xm - m^2, xm' - (m')^2\}, \quad x > 0. \quad (7.25)$$

There is also an upper bound that is uniform in B, W . If $y = \omega(Y_0)$, positivity of variance and translation invariance give

$$\omega(Y_{C_B}^2/B^6) \geq y^2, \quad -m \leq y \leq m.$$

Dropping the other nonpositive square terms shows that

$$s_{B,W}(x) \leq \max_{-m \leq y \leq m} (xy - y^2) = xm - m^2, \quad x > 2S, \quad (7.26)$$

because $m \leq S$ and $y \mapsto xy - y^2$ is increasing on $[-m, m]$ for such x . For every $x > 0$ we also have $s_{B,W}(x) \leq Sx$.

Let v_B be the harmonic function supplied by Proposition 7.1. It satisfies $v_B(z) \leq S\Re z$ on $\mathbb{H}$. At any fixed $x_0 > 2S$, (7.25) and (7.26) give $v_B(x_0) \rightarrow x_0m - m^2$. Harnack's inequality applied to the nonnegative harmonic functions $S\Re z - v_B(z)$, followed by interior compactness, supplies a subsequential harmonic limit v on $\mathbb{H}$. The same two inequalities imply $v(x) = xm - m^2$ for every $x > 2S$. The restriction of a harmonic function to the real axis is real analytic, so this identity holds for every $x > 0$.

On the other hand, (7.25) passes to this subsequence and gives $v(x) \geq xm' - (m')^2$. For $0 < x < m + m'$, these two assertions contradict

$$(xm' - (m')^2) - (xm - m^2) = (m - m')(m + m' - x) > 0.$$

Since $m > 0$, this interval is nonempty even when $m' = 0$. No smaller radius is possible, proving (7.24). $\square$

8 The total-spin limit

We now combine spatial comparison with the identification of equilibrium magnetization lengths. Fix a finite β above all the thresholds in the preceding sections and write $m = m_{S,\beta} > 0$.

Lemma 8.1. *For every finite nonempty rectangle $C \subset \mathbb{Z}^3$ and every translation-invariant equilibrium state ω ,*

$$\sum_{a=1}^3 \omega \left[\left(\frac{1}{|C|} \sum_{x \in C} S_x^a \right)^2 \right] \geq m^2. \quad (8.1)$$

Proof. In a translation-ergodic equilibrium state, translation invariance and positivity of variance give

$$\omega \left[\left(\frac{1}{|C|} \sum_{x \in C} S_x^a \right)^2 \right] \geq \omega(S_0^a)^2.$$

Summing and applying Proposition 7.3 proves the assertion for extreme equilibrium states. The equilibrium set is a compact face of the translation-invariant state space, as recalled in Section 6. By the Krein–Milman theorem it is the closed convex hull of its extreme points. The left side of (8.1) is continuous and affine in the state, so the inequality holds throughout that face. $\square$

Proposition 8.2. *The periodic zero-field Gibbs states satisfy*

$$\liminf_{L \rightarrow \infty} \frac{1}{V^2} \sum_{a=1}^3 \langle (M_L^a)^2 \rangle \geq m^2. \quad (8.2)$$

Proof. Choose a sequence of even side lengths realizing the lower limit in (8.2), and pass to a subsequence on which the Gibbs states converge locally. Their limit ω is a translation-invariant equilibrium state by Section 6. Fix the terminal scale R_* in Proposition 5.3. Its finest partition consists of rectangles with side lengths bounded above and below by constants depending on R_* only. For this fixed R_* there are finitely many possible rectangle shapes. Translation invariance and local convergence therefore imply, uniformly over all cells in that partition,

$$\liminf_{L \rightarrow \infty} \frac{1}{V} \sum_{C \in \mathcal{P}_*} |C| \sum_{a=1}^3 \left\langle \left(\frac{1}{|C|} \sum_{x \in C} S_x^a \right)^2 \right\rangle \geq m^2,$$

where Lemma 8.1 supplies the lower bound for each limiting shape. The spatial comparison (5.5), applied in each of the three directions, gives

$$\liminf_{L \rightarrow \infty} \frac{1}{V^2} \sum_{a=1}^3 \langle (M_L^a)^2 \rangle \geq m^2 - \frac{C_\beta}{\log R_*}.$$

Let $R_* \rightarrow \infty$. The sequence was chosen to realize the lower limit, so (8.2) follows along all even side lengths. $\square$

Proof of Theorem 1.1. The Hamiltonian commutes with the global spin representation of $SU(2)$. Decompose the finite-volume Hilbert space into spin- J irreducible representations and their multiplicity spaces. On each irreducible factor the Gibbs density is a scalar multiple of the identity. Thus the Gibbs trace defines a random label J_L with $0 \leq J_L \leq SV$, and

$$\sum_{a=1}^3 \langle (M_L^a)^2 \rangle = \mathbb{E}[J_L(J_L + 1)]. \quad (8.3)$$

Conditional on $J_L = J$, a component in any fixed unit direction has the uniform distribution on $-J, -J+1, \dots, J$.

We first verify the upper-tail claim (1.6). Fix $\epsilon > 0$. By the definition of the right derivative, some fixed $h > 0$ satisfies

$$p_{S,\beta}(h) - p_{S,\beta}(0) \leq h(m + \epsilon/4).$$

Since $H_{S,L}$ commutes with M_L^3 , exponential Markov and the pressure limit give

$$\begin{aligned} \limsup_{L \rightarrow \infty} \frac{1}{V} \log \mathbb{P}(M_L^3/V > m + \epsilon/2) &\leq -\beta h(m + \epsilon/2) + \beta[p_{S,\beta}(h) - p_{S,\beta}(0)] \\ &\leq -\beta h\epsilon/4 < 0. \end{aligned}$$

Here the probability is the spectral distribution of M_L^3 in the Gibbs state. If $J/V > m + \epsilon$, at least $\epsilon V/2 - 1$ of its $2J + 1$ magnetic levels exceed $(m + \epsilon/2)V$. For all sufficiently large L , their conditional probability is bounded below by a positive constant depending only on ϵ, S . Consequently

$$\mathbb{P}(J_L/V > m + \epsilon) \longrightarrow 0.$$

The laws of J_L/V are supported in the compact interval $[0, S]$. Every subsequential limit is supported in $[0, m]$ by this upper-tail bound. By (8.3), Proposition 8.2, and $J_L/V^2 \leq S/V$, its second moment is at least m^2 . A probability measure on $[0, m]$ with that property is the point mass at m . Hence

$$J_L/V \longrightarrow m \quad \text{in probability,} \quad \frac{\mathbb{E}[J_L(J_L + 1)]}{V^2} \longrightarrow m^2. \quad (8.4)$$

The second conclusion proves (1.5).

For $t \neq 0$, rotate the direction $t/|t|$ to the third axis. Conditional on $J_L = J$, the Laplace transform in (1.4) is

$$\frac{1}{2J+1} \sum_{r=-J}^J e^{|t|r/V}.$$

Whenever $J/V \rightarrow m > 0$, these Riemann sums converge to

$$\frac{1}{2} \int_{-1}^1 e^{m|t|u} du = \frac{\sinh(m|t|)}{m|t|}.$$

They are uniformly bounded by $e^{S|t|}$, so (8.4) implies convergence of their expectations. The case $t = 0$ is immediate. The normalized area measure on $\mathbb{S}^2$ has a uniform third coordinate in $[-1, 1]$, which identifies the integral in (1.4) and completes the proof. $\square$

References

- [1] Taro Asano. Theorems on the partition functions of the Heisenberg ferromagnets. *Journal of the Physical Society of Japan*, 29(2):350–359, 1970.
- [2] Jakob E. Björnberg, Jürg Fröhlich, and Daniel Ueltschi. Quantum spins and random loops on the complete graph. *Communications in Mathematical Physics*, 375:1629–1663, 2020.
- [3] Julius Borcea, Petter Brändén, and Thomas M. Liggett. Negative dependence and the geometry of polynomials. *Journal of the American Mathematical Society*, 22(2):521–567, 2009.
- [4] Joseph G. Conlon and Jan Philip Solovej. Random walk representations of the Heisenberg model. *Journal of Statistical Physics*, 64:251–270, 1991.
- [5] Michele Correggi, Alessandro Giuliani, and Robert Seiringer. Validity of the spin-wave approximation for the free energy of the Heisenberg ferromagnet. *Communications in Mathematical Physics*, 339(1):279–307, 2015.
- [6] Freeman J. Dyson, Elliott H. Lieb, and Barry Simon. Phase transitions in quantum spin systems with isotropic and nonisotropic interactions. *Journal of Statistical Physics*, 18(4):335–383, 1978.
- [7] Jürg Fröhlich, Barry Simon, and Thomas Spencer. Infrared bounds, phase transitions and continuous symmetry breaking. *Communications in Mathematical Physics*, 50(1):79–95, 1976.
- [8] Sidney Golden. Lower bounds for the Helmholtz function. *Physical Review*, 137(4B):B1127–B1128, 1965.

- [9] Daniel Kastler and Derek W. Robinson. Invariant states in statistical mechanics. *Communications in Mathematical Physics*, 3(3):151–180, 1966.
- [10] Oscar E. Lanford, III and Derek W. Robinson. Mean entropy of states in quantum-statistical mechanics. *Journal of Mathematical Physics*, 9(7):1120–1125, 1968.
- [11] Oscar E. Lanford, III and Derek W. Robinson. Statistical mechanics of quantum spin systems. III. *Communications in Mathematical Physics*, 9(4):327–338, 1968.
- [12] N. D. Mermin and H. Wagner. Absence of ferromagnetism or antiferromagnetism in one- or two-dimensional isotropic Heisenberg models. *Physical Review Letters*, 17(22):1133–1136, 1966. Erratum: *Physical Review Letters* 17 (1966), 1307.
- [13] Bruno Nachtergaele. Quasi-state decompositions for quantum spin systems. In *Probability Theory and Mathematical Statistics: Proceedings of the Sixth Vilnius Conference*, pages 565–590. VSP/TEV, Utrecht–Tokyo–Vilnius, 1994.
- [14] OpenAI. Bloch’s Law for Finite-Range Heisenberg Ferromagnets in Three Dimensions. OpenAI Math Release preprint OAI:Bloch’s-Law-for-Finite-Range-Heisenberg-Ferromagnets-in-Three-Dimensions-October-5-2026, 2026.
- [15] OpenAI. Spontaneous magnetization in the quantum Heisenberg ferromagnet. OpenAI Math Release preprint OAI:Spontaneous-magnetization-in-the-quantum-Heisenberg-ferromagnet-September-24-2026, 2026.
- [16] R. E. A. C. Paley and Antoni Zygmund. On some series of functions, (3). *Proceedings of the Cambridge Philosophical Society*, 28(2):190–205, 1932.
- [17] Robert T. Powers. Heisenberg model and a random walk on the permutation group. *Letters in Mathematical Physics*, 1(2):125–130, 1976.
- [18] David Ruelle. Extension of the Lee–Yang circle theorem. *Physical Review Letters*, 26(6):303–304, 1971. Erratum: *Physical Review Letters* 26 (1971), 870.
- [19] David Ruelle. Some remarks on the location of zeroes of the partition function for lattice systems. *Communications in Mathematical Physics*, 31:265–277, 1973.
- [20] Colin J. Thompson. Inequality with applications in statistical mechanics. *Journal of Mathematical Physics*, 6(11):1812–1813, 1965.
- [21] Bálint Tóth. Improved lower bound on the thermodynamic pressure of the spin 1/2 Heisenberg ferromagnet. *Letters in Mathematical Physics*, 28(1):75–84, 1993.
- [22] Hale F. Trotter. On the product of semi-groups of operators. *Proceedings of the American Mathematical Society*, 10(4):545–551, 1959.
- [23] Daniel Ueltschi. Universal behaviour of 3D loop soup models. In Bogdan Cichocki, Marek Napiórkowski, Jarosław Piasecki, and Piotr Szymczak, editors, *6th Warsaw School of Statistical Physics*, pages 65–100. Warsaw University Press, 2017.